The topic is

- The topic is a general or specific subject of discussion, writing, or study.
- A topic can be chosen by the speaker, writer, or student, or assigned by a teacher, editor, or supervisor.
- A topic can be broad or narrow, depending on the purpose, audience, and scope of the communication.
- A topic can be expressed as a word, phrase, question, or statement.
- A topic can be related to other topics by subtopics, categories, or themes.
- A topic can be developed by providing details, examples, evidence, or arguments.
- A topic can be organized by using an outline, a mind map, or a graphic organizer.
- A topic can be revised by adding, deleting, or rearranging information, or by changing the focus, tone, or perspective.

FUNDAMENTALS OF ELECTRONICS ENGINEERING

Electronics engineering is a branch of engineering that deals with the design, development and testing of electronic systems and products. Electronics engineers work with various components and devices that use electric current or electromagnetic fields to perform different functions.

Some of the basic concepts and topics that are essential for electronics engineering are:

- Electricity: Electricity is the flow of electric charge through a conductor. There are two types of electric current: alternating current (AC) and direct current (DC). AC current changes its direction and magnitude periodically, while DC current flows in one direction only. AC current is used for power transmission and distribution, while DC current is used for batteries, solar cells and electronic circuits.

- Circuits: A circuit is a closed path through which electric current can flow. A circuit consists of various elements such as sources, resistors, capacitors, inductors, diodes, transistors, switches, etc. that are connected by wires or traces. A circuit can be analyzed using various methods such as Kirchhoff's laws, Ohm's law, Thevenin's theorem, etc. to find the voltage, current and power across each element.

- Resistance: Resistance is a measure of how much a material opposes the flow of electric current. Resistance is measured in ohms (Ω) and depends on the material, length, cross-sectional area and temperature of the conductor. Resistance can be calculated using the formula R = L/A, where is the resistivity, L is the length and A is the cross-sectional area of the conductor. Resistance can also be measured using an ohmmeter.

- Series and parallel circuits: Series and parallel circuits are two ways of connecting circuit elements. In a series circuit, the elements are connected one after another, so that the same current flows through each element. The total resistance of a series circuit is the sum of the individual resistances. In a parallel circuit, the elements are connected across the same two points, so that the voltage across each element is the same. The total resistance of a parallel circuit is the reciprocal of the sum of the reciprocals of the individual resistances.
- Basic components: Some of the basic components that are used in electronic circuits are:
  - Resistors: Resistors are passive components that limit the flow of current and create a voltage drop in a circuit. Resistors are used for various purposes such as controlling the current, dividing the voltage, biasing the transistors, etc. Resistors have a fixed or variable resistance value that is indicated by a color code or a numerical code. Resistors can be classified into different types such as fixed, variable, potentiometer, thermistor, etc. based on their characteristics .
  - Capacitors: Capacitors are passive components that store electric charge and energy in an electric field. Capacitors consist of two conductive plates separated by a dielectric material. Capacitors have a capacitance value that is measured in farads (F) and depends on the area of the plates, the distance between them and the dielectric constant of the material. Capacitors can be classified into different types such as ceramic, electrolytic, film, etc. based on their characteristics. Capacitors are used for various purposes such as filtering, smoothing, coupling, decoupling, timing, etc. in electronic circuits .
  - Inductors: Inductors are passive components that store electric current and energy in a magnetic field. Inductors consist of a coil of wire wrapped around a core. Inductors have an inductance value that is measured in henrys (H) and depends on the number of turns, the cross-sectional area, the length and the permeability of the core. Inductors can be classified into different types such as air-core, iron-core, ferrite-core, etc. based on their characteristics. Inductors are used for various purposes such as filtering, tuning, matching, etc. in electronic circuits .
  - Diodes: Diodes are active components that allow electric current to flow in one direction only. Diodes consist of a p-n junction, which is a boundary between two types of semiconductor materials: p-type and n-type. Diodes have a forward voltage drop that is typically 0.7 V for silicon diodes and 0.3 V for germanium diodes. Diodes can be classified into different types such as rectifier, zener, light-emitting,

Unit 1 - Semiconductor Diode

- A semiconductor diode is a device that allows current to flow in one direction, but blocks it in the opposite direction.
- A semiconductor diode is made of two types of semiconductor materials: p-type and n-type. The p-type has more holes (positive charge carriers) than electrons, while the n-type has more electrons (negative charge carriers) than holes.
- The junction of the p-type and n-type materials is called the p-n junction. The p-n junction forms a depletion region, where the electrons and holes recombine and create a barrier for current flow.
- The p-type material is connected to the anode (positive terminal) of the diode, and the n-type material is connected to the cathode (negative terminal) of the diode.
- When the anode is connected to a positive voltage and the cathode is connected to a negative voltage, the diode is said to be forward biased. In this condition, the depletion region becomes narrower and the barrier is reduced. The current can flow from the anode to the cathode through the p-n junction.
- When the anode is connected to a negative voltage and the cathode is connected to a positive voltage, the diode is said to be reverse biased. In this condition, the depletion region becomes wider and the barrier is increased. The current is blocked from flowing from the anode to the cathode through the p-n junction.
- The diode has a characteristic curve that shows the relationship between the voltage across the diode and the current through the diode. The curve has a knee point, where the diode starts to conduct current in the forward bias. The knee point voltage is also called the cut-in voltage or the threshold voltage.
- The diode has a maximum reverse voltage that it can withstand before it breaks down and conducts current in the reverse bias. This voltage is also called the breakdown voltage or the peak inverse voltage.
- Some diodes have a special property that allows them to maintain a constant voltage across them when they are reverse biased. These diodes are called Zener diodes and they are used for voltage regulation.
- There are many types of diodes, such as signal diodes, rectifier diodes, switching diodes, light-emitting diodes, photodiodes, etc. Each type has a specific application and symbol .

Depletion Layer

- A depletion layer is a region in a semiconductor diode where the concentration of charge carriers is very low due to diffusion and recombination .
- A depletion layer acts as a barrier that opposes the flow of electrons from

the n-side to the p-side of the semiconductor diode .

- A depletion layer is formed when a p-type semiconductor is joined with an n-type semiconductor, creating a p-n junction .
- The formation of the depletion layer can be explained as follows :
  - At the p-n junction, there is a difference in the concentration of holes in the p-region and electrons in the n-region.
  - This causes the majority carriers (holes in the p-region and electrons in the n-region) to diffuse across the junction and recombine with the opposite carriers.
  - As a result, the p-region near the junction loses some holes and becomes negatively charged, while the n-region near the junction loses some electrons and becomes positively charged.
  - These charged regions are called the depletion regions, as they are depleted of mobile charge carriers.
  - The depletion regions extend on both sides of the junction, forming a depletion layer or a depletion zone.
  - The depletion layer creates an electric field that opposes further diffusion of charge carriers across the junction, and establishes a potential difference or a barrier potential between the p-region and the n-region.
  - The width and the potential of the depletion layer depend on the doping levels of the p-region and the n-region, and the applied voltage across the diode.
  - The depletion layer reduces the conductivity of the diode, as it has no free charge carriers to carry the current.
  - The depletion layer can be reduced or eliminated by applying a forward bias voltage across the diode, which pushes the majority carriers towards the junction and narrows the depletion layer.
  - The depletion layer can be increased or widened by applying a reverse bias voltage across the diode, which pulls the majority carriers away from the junction and expands the depletion layer.

V-I Characteristics of Semiconductor Diode

- A semiconductor diode is a device that allows current to flow in one direction but blocks it in the opposite direction.
- The V-I characteristics of a semiconductor diode is a curve that shows the relationship between the voltage across the diode and the current through it.
- The V-I characteristics of a semiconductor diode can be divided into two regions: forward bias and reverse bias.

Forward Bias

- Forward bias is the condition when the positive terminal of the battery is connected to the p-type material and the negative terminal to the n-type material of the diode.
- In forward bias, the battery voltage reduces the potential barrier at the p-n junction and allows the majority carriers to cross the junction and flow in the circuit.
- The forward current in the diode increases exponentially with the increase in the forward voltage, as shown by the curve in the figure below .
- The forward voltage required to make the diode conduct is called the cut-in voltage or the threshold voltage. It is typically 0.7 V for silicon diodes and 0.3 V for germanium diodes.
- The forward resistance of the diode is the ratio of the forward voltage to the forward current. It is very low (a few ohms) and decreases with the increase in the forward current.

Forward bias V-I characteristics of a semiconductor diode

Reverse Bias

- Reverse bias is the condition when the positive terminal of the battery is connected to the n-type material and the negative terminal to the p-type material of the diode.
- In reverse bias, the battery voltage increases the potential barrier at the p-n junction and prevents the majority carriers from crossing the junction and flowing in the circuit.
- The reverse current in the diode is very small (a few microamperes) and is due to the thermal generation of minority carriers in the diode. It is almost constant and independent of the reverse voltage, as shown by the curve in the figure below .
- The reverse voltage required to make the diode break down and conduct a large reverse current is called the breakdown voltage or the peak inverse voltage. It depends on the doping level and the structure of the diode.
- The reverse resistance of the diode is the ratio of the reverse voltage to the reverse current. It is very high (several megaohms) and increases with the increase in the reverse voltage.

Reverse bias V-I characteristics of a semiconductor diode

Ideal and Practical Diodes

- A diode is a two-terminal electronic device that allows current to flow in one direction and blocks it in the opposite direction.
- An ideal diode is a hypothetical device that has zero resistance when forward biased and infinite resistance when reverse biased.

- A practical diode is a real device that has some non-ideal characteristics, such as forward voltage drop, reverse leakage current, and breakdown voltage.

Characteristics of Ideal Diode

- An ideal diode acts as a perfect conductor when forward biased and a perfect insulator when reverse biased.
- An ideal diode has no voltage drop across it when forward biased, meaning that the voltage at the anode is equal to the voltage at the cathode.
- An ideal diode has no current flowing through it when reverse biased, meaning that the current at the anode is zero regardless of the voltage at the cathode.
- An ideal diode has a sharp transition from conducting to blocking state at zero voltage, meaning that the current-voltage (I-V) characteristic is a vertical line at zero voltage and a horizontal line at zero current.
- An ideal diode can be modeled as a switch that closes when forward biased and opens when reverse biased.

Characteristics of Practical Diode

- A practical diode acts as a non-linear resistor when forward biased and a very high resistor when reverse biased.
- A practical diode has a finite voltage drop across it when forward biased, meaning that the voltage at the anode is higher than the voltage at the cathode by a certain amount, called the forward voltage drop (Vf).
- A practical diode has a small current flowing through it when reverse biased, meaning that the current at the anode is not zero but very low, called the reverse leakage current (Ir).
- A practical diode has a gradual transition from conducting to blocking state, meaning that the current-voltage (I-V) characteristic is a curve that rises exponentially when forward biased and flattens when reverse biased.
- A practical diode can be modeled as a series combination of an ideal diode, a forward resistance (Rf), and a reverse resistance (Rr).

Comparison of Ideal and Practical Diodes

Ideal diodes	Practical diodes
Ideal diodes act as perfect conductor and perfect insulator.	Practical diodes cannot act as perfect conductor and perfect insulator.
Ideal diode draws no current when reverse biased.	Practical diode draws very low current when reverse biased.
Ideal diode offers infinite resistance when reverse biased.	Practical diode offers very high resistance when reverse biased.

Ideal diodes	Practical diodes
Ideal diode has zero voltage drop when forward biased.	Practical diode has a finite voltage drop when forward biased.
Ideal diode has a sharp transition from conducting to blocking state at zero voltage.	Practical diode has a gradual transition from conducting to blocking state.
It cannot be manufactured.	It can be manufactured.

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Diode Equivalent Circuits

- An equivalent circuit is a combination of elements that best represents the actual terminal characteristics of the device.
- An equivalent circuit can be used to simplify the analysis of a circuit containing a diode, by replacing the diode with other elements without severely affecting the behavior of the circuit.
- There are different types of equivalent circuits for a diode, depending on the level of accuracy and complexity required.
- Three models with increasing accuracy are listed below:

1. Piecewise-Linear Equivalent Circuit

- A technique for obtaining an equivalent circuit for a diode is to approximate the characteristics of the device by straight-line segments.
- The resulting equivalent circuit is naturally called the piecewise-linear equivalent circuit.
- The piecewise-linear equivalent circuit consists of a voltage source, a resistor and an ideal diode.
- The voltage source represents the forward voltage drop of the diode, which is typically 0.6 V for silicon and 0.3 V for germanium .
- The resistor represents the dynamic resistance of the diode, which is the slope of the characteristic curve at the operating point.
- The ideal diode represents the ideal behavior of the diode, which is to conduct current in one direction and block it in the other.

- The piecewise-linear equivalent circuit is useful for analyzing circuits with large variations in voltage and current, such as rectifiers and clippers.
- The piecewise-linear equivalent circuit is shown below:

Piecewise-Linear Equivalent Circuit

2. Simplified Equivalent Circuit

- The equivalent model in this case consists of a battery and an ideal diode.
- The battery represents the forward voltage drop of the diode, which is the same as in the piecewise-linear model.
- The ideal diode represents the ideal behavior of the diode, which is the same as in the piecewise-linear model.
- The simplified equivalent circuit ignores the dynamic resistance of the diode, which is assumed to be negligible.
- The simplified equivalent circuit is useful for analyzing circuits with small variations in voltage and current, such as biasing and switching circuits.
- The simplified equivalent circuit is shown below:

Simplified Equivalent Circuit

3. Ideal Diode Model

- The equivalent model in this case consists of only an ideal diode.
- The ideal diode represents the ideal behavior of the diode, which is the same as in the other models.
- The ideal diode model ignores the forward voltage drop and the dynamic resistance of the diode, which are assumed to be zero.
- The ideal diode model is useful for analyzing circuits with very small variations in voltage and current, such as logic gates and digital circuits.
- The ideal diode model is the simplest and most idealized equivalent circuit for a diode.
- The ideal diode model is shown below:

Ideal Diode Model

Zener Diodes breakdown mechanism (Zener and avalanche)

- A Zener diode is a special type of diode that can operate in the reverse breakdown region without being damaged.
- The reverse breakdown region is the region where the reverse voltage across the diode exceeds a certain value, called the breakdown voltage, and a large reverse current flows through the diode.
- The breakdown voltage of a Zener diode is determined by the doping level of the p-n junction. A higher doping level results in a lower breakdown

voltage and vice versa.

- There are two main mechanisms that cause the reverse breakdown of a Zener diode: Zener breakdown and avalanche breakdown.
- Zener breakdown occurs when the electric field across the p-n junction is so high that it breaks the covalent bonds of the semiconductor atoms and generates free electrons and holes. These charge carriers then contribute to the reverse current. Zener breakdown typically occurs at low breakdown voltages (below 5 V) and high doping levels.
- Avalanche breakdown occurs when the free electrons in the n-region gain enough kinetic energy from the electric field to collide with the semiconductor atoms and knock out more electrons. These electrons then collide with more atoms and create a chain reaction of ionization. This process also generates holes in the p-region. The electrons and holes then contribute to the reverse current. Avalanche breakdown typically occurs at high breakdown voltages (above 5 V) and low doping levels.
- Some Zener diodes can operate in both Zener and avalanche breakdown regions, depending on the applied reverse voltage. These diodes are called Zener-avalanche diodes.

Diode Application

A diode is a two-terminal electronic device that allows current to flow in only one direction. It has a low resistance in the forward direction and a high resistance in the reverse direction. Diodes are widely used in various fields of electronics for different purposes. Some of the common applications of diodes are:

- **Rectification**: A diode's most basic function is acting as a rectifier, straightening an alternating AC power source into a constant (or at least varying unidirectional) power source . Rectifiers can be classified into half-wave, full-wave, and bridge rectifiers depending on the number and arrangement of diodes used. Rectifiers are essential for converting AC power from the mains or a generator to DC power for devices such as batteries, motors, and LEDs.
- **Switching**: A diode can also act as a switch, turning on or off a circuit depending on the polarity of the input voltage . Switching diodes are fast and have a low forward voltage drop, making them suitable for high-frequency and low-power applications. Switching diodes are often used in logic circuits, pulse generators, and oscillators.
- **Source Isolation**: A diode can also isolate a source from a load, preventing reverse current from flowing back to the source . This is useful for protecting the source from damage or interference from the load. Source isolation diodes are often used in power supplies, battery chargers, and solar panels.
- **Voltage Reference**: A diode can also provide a stable and precise voltage reference for a circuit, regardless of the variations in the input voltage

or the load current . This is because a diode has a characteristic voltage drop across it in the forward direction, which depends on the type and temperature of the diode. Voltage reference diodes are often used in voltage regulators, comparators, and amplifiers.

- **Frequency Mixer**: A diode can also mix two or more input signals of different frequencies and produce an output signal that contains the sum and difference of the input frequencies . This is useful for modulating and demodulating signals in communication systems. Frequency mixer diodes are often used in radios, radars, and microwave devices.
- **Diode Detector**: A diode can also detect the presence and amplitude of an input signal by rectifying it and filtering out the unwanted components . This is useful for converting a modulated signal back to its original form in communication systems. Diode detector diodes are often used in AM radios, envelope detectors, and peak detectors.
- **Light Source**: A diode can also emit light when a current flows through it in the forward direction . This is because a diode is made of a semiconductor material that can release photons when electrons and holes recombine. Light-emitting diodes (LEDs) are efficient, durable, and versatile sources of light that can produce different colors and brightness levels. LEDs are often used in displays, indicators, lamps, and optical communication devices.
- **Temperature and Light Sensor**: A diode can also sense the temperature and light intensity of the environment by measuring the change in its electrical characteristics . This is because a diode's voltage drop, current, and resistance depend on the temperature and light exposure of the diode. Temperature and light sensor diodes are often used in thermometers, thermostats, photometers, and photovoltaic cells.
- **Solar Cell or Photovoltaic Cell**: A diode can also generate electricity when exposed to light by converting the photons into electrons and holes . This is because a diode is made of a p-n junction that can create an electric field when illuminated. Solar cells or photovoltaic cells are renewable and clean sources of energy that can power various devices and applications. Solar cells are often used in calculators, watches, satellites, and solar panels.
- **Clipper and Clamper**: A diode can also clip or clamp an input signal to a desired level by limiting or shifting its voltage range . This is useful for shaping, protecting, or modifying signals in electronic circuits. Clipper diodes are often used in

Diode Configuration

- A diode is an electrical device that allows current to flow in one direction only. It has two terminals: an anode and a cathode. The anode is the positive terminal and the cathode is the negative terminal.

- A diode is made of a semiconductor material, such as silicon, that has a p-n junction. A p-n junction is a boundary between two regions of the semiconductor that have different types of doping. Doping is the process of adding impurities to the semiconductor to change its electrical properties.
- The p-region is doped with acceptor atoms, which create holes or positive charge carriers. The n-region is doped with donor atoms, which create electrons or negative charge carriers. The p-n junction forms a depletion region, where the charge carriers are depleted due to diffusion and recombination. The depletion region creates an electric field that opposes further diffusion of charge carriers.
- A diode can be configured in different ways depending on the polarity of the applied voltage and the desired function. Some common diode configurations are:
  - Forward bias: The anode is connected to the positive terminal of the voltage source and the cathode is connected to the negative terminal. The applied voltage reduces the potential barrier of the depletion region and allows current to flow from the anode to the cathode. The diode is said to be on or conducting.
  - Reverse bias: The anode is connected to the negative terminal of the voltage source and the cathode is connected to the positive terminal. The applied voltage increases the potential barrier of the depletion region and prevents current from flowing through the diode. The diode is said to be off or non-conducting.
  - Series configuration: Two or more diodes are connected in series, with the anode of one diode connected to the cathode of the next diode. The current through the series configuration is the same for all diodes. The voltage across the series configuration is the sum of the voltages across each diode. The series configuration can be used to increase the voltage rating or the current limiting capability of the diodes.
  - Parallel configuration: Two or more diodes are connected in parallel, with the anodes of all diodes connected to one terminal of the voltage source and the cathodes of all diodes connected to the other terminal of the voltage source. The voltage across the parallel configuration is the same for all diodes. The current through the parallel configuration is the sum of the currents through each diode. The parallel configuration can be used to increase the current rating or the redundancy of the diodes.
  - Half-wave rectifier: A single diode is connected in series with a load resistor and an alternating voltage source. The diode conducts only when the input voltage is positive, and blocks the negative half-cycle of the input voltage. The output voltage across the load resistor is a pulsating direct current (DC) that has the same frequency as the input voltage. The half-wave rectifier can be used to convert alternating current (AC) to DC for low-power applications.

- Full-wave rectifier: Two or four diodes are connected in a bridge configuration with a load resistor and an alternating voltage source. The diodes conduct in pairs depending on the polarity of the input voltage, and allow both the positive and negative half-cycles of the input voltage to pass through the load resistor. The output voltage across the load resistor is a pulsating direct current (DC) that has twice the frequency of the input voltage. The full-wave rectifier can be used to convert alternating current (AC) to DC for high-power applications.
- Zener diode: A special type of diode that has a breakdown voltage that is lower than the normal reverse breakdown voltage of a diode. The breakdown voltage is the voltage at which the diode starts to conduct in the reverse direction. The zener diode can be used as a voltage regulator, as it maintains a constant voltage across itself when the input voltage is higher than the breakdown voltage.
- Light-emitting diode (LED): A special type of diode that emits light when current flows through it in the forward direction. The color of the light depends on the band gap of the semiconductor material used to make the diode. The LED can be used as an indicator, a display, or a source of illumination.

Half and Full Wave Rectification

Introduction

Rectification is the process of converting an alternating current (AC) into a direct current (DC) by using one or more diodes. A diode is a semiconductor device that allows current to flow in one direction only. Rectification is important for many applications that require a steady and constant DC voltage, such as power supplies, battery chargers, LED lights, etc.

Half Wave Rectification

- A half wave rectifier is a rectifier that uses only one diode to convert one half cycle of the AC input into a pulsating DC output.
- The other half cycle of the AC input is blocked by the diode and does not contribute to the output.
- The output voltage and current are proportional to the input voltage and current, but only for the positive half cycle.
- The output frequency is the same as the input frequency, but the output has a lot of ripple, which is the variation of the output voltage around the average value.
- The advantage of a half wave rectifier is its simplicity and low cost, but the disadvantage is its low efficiency and high ripple.

Half wave rectifier circuit and output waveform

Full Wave Rectification

- A full wave rectifier is a rectifier that uses two or four diodes to convert both half cycles of the AC input into a pulsating DC output.
- There are two types of full wave rectifiers: center-tapped and bridge.
- A center-tapped full wave rectifier uses a transformer with a center-tapped secondary winding and two diodes to rectify the AC input. The center tap provides a common ground for the two diodes, which conduct alternately during each half cycle of the input. The output voltage and current are twice that of a half wave rectifier, but the output frequency is also doubled.
- A bridge full wave rectifier uses four diodes arranged in a bridge configuration to rectify the AC input. No transformer is required, but the diodes must have a higher voltage rating than the input. The output voltage and current are the same as that of a center-tapped full wave rectifier, but the output frequency is also the same as the input frequency.

Center-tapped and bridge full wave rectifier circuits and output waveforms

Comparison

- The main difference between half wave and full wave rectifiers is their efficiency. Efficiency is the ratio of the output power to the input power. A half wave rectifier has a low efficiency of about 40.6%, while a full wave rectifier has a high efficiency of about 81.2%.
- Another difference is the ripple factor, which is the ratio of the root mean square (RMS) value of the ripple voltage to the DC value of the output voltage. A lower ripple factor means a smoother output voltage. A half wave rectifier has a high ripple factor of about 1.21, while a full wave rectifier has a low ripple factor of about 0.482.
- A third difference is the transformer utilization factor (TUF), which is the ratio of the output power to the AC rating of the transformer. A higher TUF means a better utilization of the transformer. A half wave rectifier has a low TUF of about 0.287, while a full wave rectifier has a high TUF of about 0.693.

Summary

- Rectification is the process of converting AC into DC by using diodes.
- A half wave rectifier uses one diode to convert one half cycle of the AC input into a pulsating DC output. It has a low efficiency, high ripple, and low TUF.
- A full wave rectifier uses two or four diodes to convert both half cycles of the AC input into a pulsating DC output. It has a high efficiency, low ripple, and high TUF.
- A full wave rectifier is preferred over a half wave rectifier for most applications that require a steady and constant DC voltage.

Clippers for the notes of the Unit 1 - Semiconductor Diode in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING

- A **semiconductor diode** is a device that allows current to flow in one direction, but blocks it in the opposite direction .
- A semiconductor diode is made of two types of semiconductor materials: **p-type** and **n-type**. The p-type has a surplus of positive charges (holes), while the n-type has a surplus of negative charges (electrons) .
- The junction of the p-type and n-type materials is called the **pn junction**. The pn junction has two terminals: the **anode** (connected to the p-type) and the **cathode** (connected to the n-type) .
- When the anode is connected to a positive voltage and the cathode is connected to a negative voltage, the diode is said to be **forward biased**. In this condition, the diode allows current to flow from the anode to the cathode .
- When the anode is connected to a negative voltage and the cathode is connected to a positive voltage, the diode is said to be **reverse biased**. In this condition, the diode blocks current from flowing from the anode to the cathode, except for a very small **leakage current** .
- The **characteristic curve** of a diode shows the relationship between the voltage across the diode and the current through the diode. The curve has two regions: the **forward region** and the **reverse region** .
- In the forward region, the current increases exponentially with the voltage, until it reaches a **saturation current**. The voltage required to make the diode conduct is called the **threshold voltage** or the **cut-in voltage** .
- In the reverse region, the current is almost zero, until the voltage reaches a **breakdown voltage**. The breakdown voltage is the voltage at which the diode starts to conduct in the reverse direction, due to the **avalanche effect** or the **Zener effect** .
- A **clipper** is a circuit that uses one or more diodes to limit the voltage of an input signal to a certain range. A clipper can be used to protect a circuit from overvoltage, to remove unwanted parts of a signal, or to shape a signal into a desired form .
- There are different types of clippers, such as **series clippers** and **parallel clippers**, **positive clippers** and **negative clippers**, **biased clippers** and **unbiased clippers**, and **combinational clippers** .
- A series clipper has the diode connected in series with the load, while a parallel clipper has the diode connected in parallel with the load .
- A positive clipper removes the positive parts of the input signal that are above a certain level, while a negative clipper removes the negative parts of the input signal that are below a certain level .
- A biased clipper has a dc voltage source connected to the diode, which shifts the clipping level from zero to a desired value, while an unbiased clipper has no dc voltage source connected to the diode, which clips the

signal at zero .
- A combinational clipper has more than one diode and/or more than one dc voltage source, which allows clipping the signal at different levels for different parts of the signal .

Clampers

- Clampers are electronic circuits that shift the dc level of the AC signal .
- Clampers are also known as DC voltage restorers or level shifter.
- Clampers are used to add the dc level to the ac input signal. The input swing of a waveform is equal to the output swing.
- Clampers can be classified as positive or negative, and biased or unbiased.
- A positive clamper circuit (negative peak clamper) outputs a purely positive waveform from an input signal; it offsets the input signal so that all of the waveform is greater than 0 V.
- A negative clamper circuit (positive peak clamper) outputs a purely negative waveform from an input signal; it offsets the input signal so that all of the waveform is less than 0 V.
- A biased clamper circuit adds a fixed dc voltage to the input signal, either positive or negative, depending on the polarity of the bias.
- An unbiased clamper circuit does not add any dc voltage to the input signal, but only shifts it up or down.
- A clamper circuit consists of a diode, a capacitor, and a resistor.
- The diode determines the polarity of the clamping, and the capacitor determines the amount of the clamping.
- The resistor provides a discharge path for the capacitor and limits the current through the diode.
- A clamper circuit can be used as a DC restorer in composite video circuitry in both television transmitters and receivers.
- A clamper circuit can also be used to protect a circuit from overvoltage or undervoltage conditions.

Zener Diode as Shunt Regulator

- A Zener diode is a special type of diode that can operate in the reverse breakdown region, where it maintains a nearly constant voltage across its terminals.
- A Zener diode can be used as a shunt voltage regulator, which is a device that regulates the output voltage across a load by shunting excess current to the ground.
- The basic circuit of a Zener diode shunt regulator is shown below:

Zener diode shunt regulator circuit

- In this circuit, the Zener diode is connected in parallel with the load resistor RL, and the input voltage Vin is applied through a series resistor RS.

- The operation of the Zener diode shunt regulator is as follows:
  - When Vin is less than the Zener breakdown voltage VZ, the Zener diode is in the reverse cutoff region, and no current flows through it. The output voltage Vout is equal to Vin minus the voltage drop across RS.
  - When Vin is equal to or greater than VZ, the Zener diode is in the reverse breakdown region, and a current IZ flows through it. The output voltage Vout is equal to VZ, which is the Zener voltage. The excess current IS - IL - IZ is shunted to the ground through the Zener diode.
- The advantages of using a Zener diode as a shunt regulator are:
  - It provides a better regulation over a wide range of load currents and input voltages.
  - It has a higher current capability than a series regulator.
  - It is very economical and simple to implement.
- The disadvantages of using a Zener diode as a shunt regulator are:
  - It has a poor efficiency, as the excess current is wasted as heat in the Zener diode and the series resistor.
  - It has a limited power rating, as the Zener diode and the series resistor have to dissipate the heat generated by the shunt current.
  - It has a high output impedance, as the Zener diode is in parallel with the load resistor.

Voltage-Multiplier Circuits

- A voltage multiplier is an electronic circuit consisting of capacitors and diodes and is used to multiply or rise the voltage level of an AC signal.
- The voltage multiplier receives an AC voltage of lower value, converts it into a DC voltage and increases its voltage level.
- Voltage multipliers are classified as voltage doublers, triplers, quadruplers, etc, depending on the ratio of the output voltage to the input voltage.
- In theory, any desired amount of voltage multiplication can be obtained by cascading voltage doublers, but in practice, the output voltage drops due to the losses in the circuit components.
- Voltage multipliers can be used to generate a few volts for electronic appliances, to millions of volts for purposes such as high-energy physics experiments and lightning safety testing.

Voltage Doubler Circuit

- A voltage doubler is a voltage multiplier circuit that produces an output voltage that is twice the peak input voltage of an AC signal.
- There are two types of voltage doubler circuits: half-wave and full-wave.

- A half-wave voltage doubler consists of two capacitors and two diodes connected as shown below:

Half-wave voltage doubler circuit

- The operation of the half-wave voltage doubler can be explained as follows:
  - During the positive half-cycle of the input AC signal, diode D1 is forward biased and capacitor C1 is charged to the peak input voltage Vp.
  - During the negative half-cycle of the input AC signal, diode D2 is forward biased and capacitor C2 is charged to the peak input voltage Vp in series with capacitor C1, resulting in an output voltage of 2Vp across C2.
  - The output voltage is a pulsating DC voltage with a ripple frequency equal to the input frequency.
- A full-wave voltage doubler consists of four diodes and two capacitors connected as shown below:

Full-wave voltage doubler circuit

- The operation of the full-wave voltage doubler can be explained as follows:
  - During the positive half-cycle of the input AC signal, diode D1 is forward biased and capacitor C1 is charged to the peak input voltage Vp, while diode D3 is reverse biased and capacitor C2 is disconnected from the circuit.
  - During the negative half-cycle of the input AC signal, diode D2 is forward biased and capacitor C2 is charged to the peak input voltage Vp in series with capacitor C1, resulting in an output voltage of 2Vp across C2, while diode D4 is reverse biased and capacitor C1 is disconnected from the circuit.
  - The output voltage is a pulsating DC voltage with a ripple frequency equal to twice the input frequency.
- The advantages of a full-wave voltage doubler over a half-wave voltage doubler are:
  - Higher output voltage due to lower voltage drop across the diodes.
  - Lower output ripple due to higher ripple frequency.
  - Higher output current due to lower output impedance.

Voltage Tripler Circuit

- A voltage tripler is a voltage multiplier circuit that produces an output voltage that is three times the peak input voltage of an AC signal.
- A voltage tripler circuit can be constructed by adding a capacitor and a diode to a half-wave voltage doubler circuit as shown below:

Voltage tripler circuit

- The operation of the voltage tripler circuit can be explained as follows:

- During the positive half-cycle of the input AC signal, diode D1 is forward biased and capacitor C1 is charged to the peak input voltage Vp, while diode D2 and D3 are reverse biased and capacitors C2 and C3 are disconnected from the circuit.
- During the negative half-cycle of the input AC signal, diode D2 is forward biased and capacitor C2 is charged to the peak input voltage Vp in series with capacitor C1, resulting in a voltage of 2Vp across C2, while diode D3 is forward biased and capacitor C3 is charged to the peak input voltage Vp in

Special Purpose Two Terminal Devices

Two terminal devices are electronic components that have only two terminals (anode and cathode) and allow current to flow only in one direction. The most common two terminal device is the diode, which is a semiconductor device that has a PN junction. Diodes have many applications in electronics, such as rectification, switching, voltage regulation, signal modulation, etc.

However, there are some special types of diodes that have different characteristics and functions than the ordinary diodes. These are called special purpose two terminal devices, and they include:

- **Tunnel diode**: A tunnel diode is a diode that has a very thin and highly doped PN junction, which allows quantum mechanical tunneling of electrons across the junction. This results in a negative resistance region in the current-voltage characteristic, where the current decreases as the voltage increases. Tunnel diodes can operate at very high frequencies (up to several hundred GHz) and are used for microwave oscillators, amplifiers, and switches.

- **Photo diode**: A photo diode is a diode that is sensitive to light. When light falls on the PN junction, it generates electron-hole pairs, which increase the current flow. Photo diodes can be used as light detectors, optical receivers, solar cells, etc. Photo diodes can be reverse biased or forward biased, depending on the application.

- **Varactor diode**: A varactor diode is a diode that has a variable capacitance, which depends on the applied reverse bias voltage. The PN junction acts as a capacitor, and the width of the depletion region changes with the voltage. Varactor diodes can be used for tuning circuits, frequency modulation, voltage controlled oscillators, etc.

- **Schottky diode**: A Schottky diode is a diode that has a metal-semiconductor junction, instead of a PN junction. The metal acts as the anode, and the semiconductor acts as the cathode. Schottky diodes have a lower forward voltage drop (about 0.3 V) and a faster switching speed than ordinary diodes. Schottky diodes can be used for power rectification, logic circuits, voltage clamping, etc.

- **Light emitting diode (LED)**: A light emitting diode is a diode that emits light when forward biased. The light is produced by the recombination of electrons and holes in the semiconductor material, which releases energy in the form of photons. The color of the light depends on the band gap of the material. LEDs can be used for indicators, displays, lighting, optical communication, etc.
- **Silicon controlled rectifier (SCR)**: A silicon controlled rectifier is a device that has three terminals (anode, cathode, and gate) and four layers of semiconductor material (PNPN). It acts as a switch that can be turned on by a small gate current, and turned off by reducing the anode current below a certain level. SCRs can be used for power control, motor speed control, switching, etc.

These are some of the special purpose two terminal devices that are used in electronics engineering. They have different characteristics and functions than the ordinary diodes, and they can be used for various applications.

Light-Emitting Diodes

- A light-emitting diode (LED) is a semiconductor device that emits light when current flows through it.
- The light is produced by the recombination of electrons and electron holes in the semiconductor, a process called “electroluminescence” .
- The wavelength of the light emitted depends on the energy band gap of the semiconductors used. Different colors of light can be obtained by using different materials or doping agents.
- LEDs are p-n junction devices made from extrinsic semiconductors. An n-type and a p-type semiconductor are put in contact with each other to form a p-n junction diode.
- LEDs allow the current to flow in the forward direction and block the current in the reverse direction, like a normal diode.
- When a forward bias voltage is applied to the LED, electrons from the n-type region cross the depletion layer and recombine with holes from the p-type region, releasing energy in the form of photons .
- The color of the light depends on the energy difference between the conduction band and the valence band of the semiconductor, which is determined by the material and the doping level .
- Some common materials used for LEDs are gallium arsenide (GaAs), gallium phosphide (GaP), gallium arsenide phosphide (GaAsP), indium gallium nitride (InGaN), and aluminum gallium indium phosphide (AlGaInP) .
- LEDs have many advantages over conventional light sources, such as low power consumption, long lifetime, high efficiency, small size, fast switching,

and environmental friendliness .

- LEDs have many applications in various fields, such as displays, indicators, lighting, communication, sensors, and optical devices .

Photo Diodes

- A photo diode is a light-sensitive semiconductor diode that produces current when it absorbs photons .
- A photo diode is designed to operate in reverse bias, meaning that the anode is connected to the negative terminal and the cathode is connected to the positive terminal of a voltage source.
- A photo diode has a nearly linear relationship of current to received optical power, meaning that the more light falls on the device, the more current flows through it.
- A photo diode can be used to measure light intensity, either for its own sake or as a measure of some other property (such as smoke density, radiation level, etc.).
- A photo diode can also be used to generate electric power from solar radiation, in which case it is called a solar cell.
- A photo diode consists of a p-n junction, where the p-type material is exposed to light and the n-type material is connected to the circuit.
- When light photons with sufficient energy strike the p-type material, they create electron-hole pairs that diffuse across the junction and contribute to the reverse current.
- The reverse current of a photo diode is proportional to the light intensity and the area of the device.
- A photo diode can be packaged with lenses or optical filters to focus or filter the incoming light, depending on the application .
- A photo diode can have different spectral responses, depending on the material and the wavelength of the light.
- Some common materials used for photo diodes are silicon, germanium, gallium arsenide, indium gallium arsenide, and lead sulfide.
- A photo diode can have different modes of operation, such as photovoltaic mode, photoconductive mode, or avalanche mode, depending on the bias voltage and the circuit configuration .

Varactor Diodes

- A varactor diode is a type of diode that acts as a variable capacitor when reverse biased .
- The capacitance of a varactor diode depends on the applied reverse voltage and the physical characteristics of the diode .
- The symbol of a varactor diode is shown below :

varactor diode symbol

- Varactor diodes are mainly used in applications that require tuning or frequency control, such as voltage-controlled oscillators, frequency multipliers, parametric amplifiers, and filters .
- The advantages of varactor diodes are that they are small, cheap, reliable, and can be easily controlled by varying the voltage .
- The disadvantages of varactor diodes are that they have high series resistance, low Q factor, and non-linear capacitance-voltage characteristics .
- The capacitance of a varactor diode can be calculated by the following formula :

varactor diode capacitance formula

where C is the capacitance, C0 is the capacitance at zero bias, V is the reverse voltage, and n is a constant that depends on the doping profile of the diode.

- Some examples of varactor diodes are Diodes Inc. SMV1232-079LF, Infineon BBY40, NXP BB833, ON Semiconductor MV2109, Skyworks SMV1405, Toshiba 1SV149, and Microchip MA4E2054 .

Tunnel Diodes

- A tunnel diode is a type of semiconductor diode that has effectively "negative resistance" due to the quantum mechanical effect called tunneling.
- Tunneling is the phenomenon where an electron can pass through a potential barrier that is higher than its kinetic energy.
- A tunnel diode is made of a heavily doped p-n junction that is about 10 nm wide. The heavy doping results in a broken band gap, where conduction band electron states on the n-side are aligned with valence band hole states on the p-side.
- The symbol of a tunnel diode is shown below:

Tunnel diode symbol

- The current-voltage (I-V) characteristic of a tunnel diode is shown below:

Tunnel diode I-V characteristic

- The I-V characteristic has three regions: forward bias, negative resistance, and reverse bias.
- In the forward bias region, the current increases rapidly as the voltage increases until it reaches a peak value (Ip) at a voltage (Vp). This is due to the tunneling of electrons from the valence band of the p-side to the conduction band of the n-side.
- In the negative resistance region, the current decreases as the voltage increases until it reaches a valley value (Iv) at a voltage (Vv). This is due to the depletion of available states for tunneling as the voltage increases.

- In the reverse bias region, the current increases slowly as the voltage increases until it reaches a breakdown value (Ib) at a voltage (Vb). This is due to the conventional diode behavior of the p-n junction.
- The negative resistance region of the tunnel diode makes it useful for high-frequency applications, such as oscillators, amplifiers, and switches .
- Some advantages of tunnel diodes are: high speed, low noise, low power consumption, and simple circuitry .
- Some disadvantages of tunnel diodes are: low output voltage, low dynamic range, temperature sensitivity, and fabrication difficulty .
- Some applications of tunnel diodes are: microwave oscillators, frequency converters, logic circuits, pulse generators, and memory devices .

Unit 2 - Bipolar Junction Transistor

- A bipolar junction transistor (BJT) is a type of semiconductor device that can amplify or switch electrical signals.
- A BJT consists of three regions: the emitter, the base, and the collector. The emitter and the collector are doped with the same type of impurities (either n-type or p-type), while the base is doped with the opposite type of impurities. The base region is very thin and lightly doped compared to the emitter and the collector.
- A BJT can operate in three modes: cut-off, active, and saturation. In the cut-off mode, the base-emitter junction is reverse biased and the base-collector junction is either reverse biased or weakly forward biased. There is no current flow in the transistor. In the active mode, the base-emitter junction is forward biased and the base-collector junction is reverse biased. The current flow in the transistor is proportional to the base current and the current gain of the transistor. In the saturation mode, both the base-emitter and the base-collector junctions are forward biased. The current flow in the transistor is limited by the external circuit and the transistor acts as a closed switch.
- A BJT can be classified into two types: npn and pnp. In an npn transistor, the emitter and the collector are n-type regions and the base is a p-type region. The conventional current flows from the emitter to the collector through the base. In a pnp transistor, the emitter and the collector are p-type regions and the base is an n-type region. The conventional current flows from the collector to the emitter through the base.
- A BJT can be used as an amplifier or a switch in various circuits. Some common applications of BJTs are audio amplifiers, oscillators, logic gates, voltage regulators, and power converters.

Transistor Construction

- A transistor is a three-layer semiconductor device that can amplify or switch electric currents.

- A transistor consists of two types of semiconductor materials: P-type and N-type. P-type has more holes (positive charge carriers) than electrons, while N-type has more electrons (negative charge carriers) than holes.
- A transistor can be either PNP or NPN, depending on the arrangement of the layers. In a PNP transistor, a thin layer of N-type material is sandwiched between two thicker layers of P-type material. In an NPN transistor, a thin layer of P-type material is sandwiched between two thicker layers of N-type material .
- The three layers of a transistor are called the emitter, the base, and the collector. The emitter is the layer that supplies the input current to the transistor. The base is the layer that controls the output current of the transistor. The collector is the layer that receives the output current from the transistor .
- The emitter and the collector are usually doped more heavily than the base, meaning they have more excess charge carriers. The base is usually very thin and lightly doped, meaning it has fewer charge carriers.
- The transistor works by using the base-emitter junction as a diode that controls the current flow through the collector-emitter junction. When the base-emitter junction is forward biased (positive voltage on the P-type side and negative voltage on the N-type side), the current flows from the emitter to the base and from the base to the collector. When the base-emitter junction is reverse biased (negative voltage on the P-type side and positive voltage on the N-type side), the current is blocked from the emitter to the base and from the base to the collector.
- The amount of current that flows from the emitter to the collector depends on the voltage applied to the base-emitter junction. A small change in the base voltage can cause a large change in the collector current, thus amplifying the input signal. Alternatively, the base voltage can be used to switch the collector current on or off, thus acting as a switch.
- The symbols for PNP and NPN transistors are shown below. The arrow on the emitter indicates the direction of the conventional current flow when the transistor is in active mode (base-emitter junction forward biased and collector-emitter junction reverse biased).

PNP and NPN transistor symbols

Operation of Bipolar Junction Transistor

- A bipolar junction transistor (BJT) is a type of transistor that uses both electron and hole charge carriers.
- A BJT has three terminals: base (B), collector (C) and emitter (E).
- A BJT can be either npn or pnp, depending on the arrangement of the n-type and p-type semiconductor materials.
- A BJT can operate in three regions: active, saturation and cutoff.
- In the active region, the base-emitter junction is forward biased and the

base-collector junction is reverse biased. The BJT acts as an amplifier, where a small change in the base current causes a large change in the collector current.
- In the saturation region, both the base-emitter and the base-collector junctions are forward biased. The BJT acts as a switch, where the collector current is maximum and independent of the base current.
- In the cutoff region, both the base-emitter and the base-collector junctions are reverse biased. The BJT acts as a switch, where the collector current is zero and independent of the base current.
- The operation of a BJT can be explained by using the concept of minority carrier injection and diffusion.
- In an npn BJT, when the base-emitter junction is forward biased, electrons are injected from the n-type emitter into the p-type base. These electrons are the minority carriers in the base region, and they diffuse across the base towards the base-collector junction.
- When the base-collector junction is reverse biased, the electrons that reach the junction are swept into the n-type collector by the electric field. These electrons form the collector current, which is proportional to the base current.
- In a pnp BJT, the operation is similar, but the roles of electrons and holes are reversed. Holes are injected from the p-type emitter into the n-type base, and they diffuse across the base towards the base-collector junction. The holes that reach the junction are swept into the p-type collector by the electric field, forming the collector current.

Amplification action for the notes of the Unit 2 - Bipolar Junction Transistor in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING

- A bipolar junction transistor (BJT) is a three-terminal device that can amplify the input signal and produce an output signal that is proportional to the input signal.
- A BJT consists of two p-n junctions, one between the base and the emitter, and the other between the base and the collector.
- The base is the control terminal, the emitter is the input terminal, and the collector is the output terminal.
- A BJT can be operated in three regions: cut-off, active, and saturation.
- In the cut-off region, the base-emitter junction and the base-collector junction are both reverse biased, and no current flows through the transistor.
- In the saturation region, the base-emitter junction and the base-collector junction are both forward biased, and the transistor acts as a closed switch.
- In the active region, the base-emitter junction is forward biased and the

base-collector junction is reverse biased, and the transistor acts as an amplifier.

- The amplification action of a BJT is based on the fact that a small change in the base current can cause a large change in the collector current, as the collector current is equal to the product of the base current and the current gain.
- The current gain, also known as the beta ( ) or the common-emitter current gain, is the ratio of the collector current to the base current.
- The voltage gain, also known as the common-emitter voltage gain, is the ratio of the output voltage across the collector and the emitter to the input voltage across the base and the emitter.
- The power gain, also known as the common-emitter power gain, is the product of the current gain and the voltage gain.
- Depending on the quantity amplified by the circuit, the BJT amplifier can be a voltage amplifier, a current amplifier, or a power amplifier.
- A BJT amplifier can be configured in three ways: common-emitter, common-base, and common-collector.
- In the common-emitter configuration, the emitter is the common terminal, the base is the input terminal, and the collector is the output terminal.
- In the common-base configuration, the base is the common terminal, the emitter is the input terminal, and the collector is the output terminal.
- In the common-collector configuration, the collector is the common terminal, the base is the input terminal, and the emitter is the output terminal.
- Each configuration has different characteristics, such as input impedance, output impedance, voltage gain, current gain, and phase shift.
- A BJT amplifier can be used for various applications, such as audio amplification, signal processing, switching, and logic circuits.

Common Base Configuration of BJT

- The common base configuration is one of the three basic ways to connect a bipolar junction transistor (BJT) as an amplifier.
- In this configuration, the base terminal of the BJT is a common terminal to both the input and output signals, hence its name common base (CB).
- The input signal is applied between the emitter and the base, and the output signal is taken from the collector and the base.
- The common base configuration is less common as an amplifier than compared to the more popular common emitter (CE) or common collector (CC) configurations, but it is still used due to its unique input/output characteristics.
- Some of the advantages of the common base configuration are:
  - It has a high voltage gain, which is the ratio of output voltage to input voltage.
  - It has a high input resistance and a low output resistance, which makes it suitable for impedance matching.

  - It has a high frequency response, which means it can amplify high-frequency signals without much attenuation or distortion.
  - It has a low noise figure, which means it produces less noise in the output signal than other configurations.
- Some of the disadvantages of the common base configuration are:
  - It has a low current gain, which is the ratio of output current to input current.
  - It has a low power gain, which is the product of voltage gain and current gain.
  - It has a low input impedance, which means it requires a large input signal to produce a significant output signal.
  - It has a high output impedance, which means it cannot drive a low-resistance load without losing voltage.
- The common base configuration can be analyzed using the following formulas and equations :
  - The current gain, alpha ( ), is the ratio of collector current (IC) to emitter current (IE):
    - = IC / IE
  - The voltage gain, AV, is the ratio of output voltage (Vout) to input voltage (Vin):
    - AV = Vout / Vin = - * (RC / RE)
  - The input resistance, Rin, is the ratio of input voltage (Vin) to input current (Iin):
    - Rin = Vin / Iin = (1 +) * (VT / IE)
  - The output resistance, Rout, is the ratio of output voltage (Vout) to output current (Iout):
    - Rout = Vout / Iout = RC
  - The power gain, AP, is the product of voltage gain and current gain:
    - AP = AV *
  - Where VT is the thermal voltage, which is approximately 26 mV at room temperature.

Common Emitter

- A common emitter (CE) amplifier is one of the three basic single-stage bipolar junction transistor (BJT) amplifier topologies, typically used as a voltage amplifier .
- A BJT is a type of transistor that uses both electrons and holes as charge carriers.
- In a CE amplifier, the emitter terminal of the BJT is common to both the input and output circuits .
- The input signal is applied at the base terminal and the output signal is obtained at the collector terminal .
- The base current controls the collector current, and the ratio of change in collector current to change in base current is defined as the current gain

of the CE transistor.

- The emitter current is the sum of the base current and the collector current.
- The CE amplifier offers high current gain, medium input resistance and high output resistance .
- The CE amplifier also has a high voltage gain, which is the ratio of change in output voltage to change in input voltage.
- The CE amplifier requires proper biasing to operate in the active region, where the transistor acts as a linear amplifier.
- The CE amplifier can be analyzed using the hybrid-pi model, which is a small-signal equivalent circuit that represents the transistor behavior at low frequencies.

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https://en.wikipedia.org/wiki/Bipolar_junction_transistor

Common Collector Configuration

- Common collector configuration is one of the three basic single-stage bipolar junction transistor (BJT) amplifier topologies, along with common emitter and common base configurations.
- In common collector configuration, the base terminal of the transistor serves as the input, the emitter terminal is the output and the collector terminal is common for both input and output. Hence, it is named as common collector configuration .
- The collector terminal is grounded so the common collector configuration is also known as grounded collector configuration. Sometimes common collector configuration is also referred to as emitter follower, voltage follower, common collector amplifier, CC amplifier, or CC configuration .
- The common collector amplifier is often used as a voltage buffer, because it has a high input impedance, a low output impedance and a non-inverting voltage gain of approximately one .
- The input circuit is connected between emitter and base and the output is taken from the collector and emitter. The input signal is applied to the base-emitter junction and the output signal is taken from the emitter-collector junction.
- The common collector amplifier has the following characteristics :
  - Voltage gain: The voltage gain is the ratio of the output voltage to the input voltage. The voltage gain of the common collector amplifier is slightly less than one, because the output voltage is the input voltage minus the base-emitter voltage drop. The voltage gain is given by:

`A_v = \frac{V_o}{V_i} = \frac{V_e}{V_b} = \frac{R_e}{R_e + r_e} \approx 1`
where `V_o` is the output voltage, `V_i` is the input voltage, `V_e` is the emitter voltage, `V_b` is the base voltage, `R_e` is the external emitter resistance, and `r_e` is the internal emitter resistance.

  - Current gain: The current gain is the ratio of the output current to the input current. The current gain of the common collector amplifier is equal to the current gain of the transistor, which is typically very high. The current gain is given by:

`A_i = \frac{I_o}{I_i} = \frac{I_e}{I_b} = \beta`
where `I_o` is the output current, `I_i` is the input current, `I_e` is the emitter current, `I_b` is the base current, and is the current gain of the transistor.

  - Power gain: The power gain is the ratio of the output power to the input power. The power gain of the common collector amplifier is the product of the voltage gain and the current gain. The power gain is given by:

`A_p = \frac{P_o}{P_i} = A_v \times A_i = \beta \times \frac{R_e}{R_e + r_e} \approx \be`
where `P_o` is the output power and `P_i` is the input power.

  - Input impedance: The input impedance is the ratio of the input voltage to the input current. The input impedance of the common collector amplifier is very high, because the input current is very small compared to the output current. The input impedance is given by:

`Z_i = \frac{V_i}{I_i} = \frac{V_b}{I_b} = (\beta + 1) \times r_e`
where `Z_i` is the input impedance.

  - Output impedance: The output impedance is the ratio of the output voltage to the output current. The output impedance of the common collector amplifier is very low, because the output voltage is very close to the input voltage. The output impedance is given by:

`Z_o = \frac{V_o}{I_o} = \frac{V_e}{I_e} = \frac{R_e}{\beta + 1}`
where `Z_o` is the output impedance.

- The common collector amplifier has the following advantages :
  - It provides a high input impedance and a low output impedance, which makes it suitable for impedance matching and voltage buffering applications.
  - It has a non-inverting voltage gain of approximately one, which means it does not alter the phase or amplitude of the input signal.

Unit 3 - Field Effect Transistor

- A field effect transistor (FET) is a type of transistor that uses an electric field to control the flow of current in a semiconductor.
- FETs have three terminals: source, gate, and drain.
- The source is the terminal where the current enters the device, the drain is the terminal where the current leaves the device, and the gate is the terminal that controls the current flow by applying a voltage.

- There are two main types of FETs: junction field effect transistors (JFETs) and metal-oxide-semiconductor field effect transistors (MOSFETs).
- JFETs are FETs that use a pn junction as the gate, and have two modes of operation: depletion mode and enhancement mode.
- In depletion mode, the JFET is normally on, and the gate voltage can reduce the current flow by creating a depletion region that narrows the channel.
- In enhancement mode, the JFET is normally off, and the gate voltage can increase the current flow by creating an inversion layer that widens the channel.
- MOSFETs are FETs that use a metal-oxide layer as the gate, and have four modes of operation: enhancement mode, depletion mode, n-channel, and p-channel.
- In enhancement mode, the MOSFET is normally off, and the gate voltage can increase the current flow by creating an inversion layer that forms a channel between the source and the drain.
- In depletion mode, the MOSFET is normally on, and the gate voltage can reduce the current flow by creating a depletion region that reduces the channel width.
- In n-channel mode, the MOSFET uses n-type semiconductor as the channel, and the gate voltage is positive with respect to the source.
- In p-channel mode, the MOSFET uses p-type semiconductor as the channel, and the gate voltage is negative with respect to the source.
- FETs are widely used in electronic circuits for amplification, switching, modulation, and sensing applications.
- FETs have some advantages over bipolar junction transistors (BJTs), such as higher input impedance, lower power consumption, and better thermal stability.
- FETs also have some disadvantages, such as lower transconductance, higher noise, and susceptibility to damage by electrostatic discharge.

Construction and Characteristic of JFETs

- A JFET (Junction Field Effect Transistor) is a three-terminal semiconductor device that can be used as a voltage-controlled resistor, switch, or amplifier .
- A JFET is constructed using a long channel of semiconductor material, either N-type or P-type, with two ohmic contacts at the ends, called the source (S) and the drain (D), and a reverse-biased PN junction on each side, called the gate (G) .
- The symbols and basic construction for both N-channel and P-channel JFETs are shown below:

JFET symbols and construction

- The operation of a JFET is based on the modulation of the channel resistance by the gate voltage. When the gate voltage is zero, the channel is fully open and the JFET conducts the maximum current from source to drain, called the drain saturation current (IDSS) .

- When the gate voltage is negative for an N-channel JFET, or positive for a P-channel JFET, the depletion regions around the PN junctions expand, reducing the effective width of the channel and increasing its resistance. This reduces the drain current for a given drain voltage .
- When the gate voltage reaches a certain value, called the pinch-off voltage (VP), the depletion regions touch each other and the channel is closed, resulting in zero drain current. The pinch-off voltage is negative for an N-channel JFET, and positive for a P-channel JFET .
- The relationship between the drain current and the gate voltage is called the transfer characteristic of the JFET. It is usually expressed as a function of the normalized drain current (ID/IDSS) and the normalized gate voltage (VGS/VP) .
- The transfer characteristic of a JFET is shown below:

JFET transfer characteristic

- The relationship between the drain current and the drain voltage is called the drain characteristic of the JFET. It depends on the gate voltage and the channel resistance .
- The drain characteristic of a JFET has two regions: the ohmic region, where the drain current is linearly proportional to the drain voltage, and the saturation region, where the drain current is independent of the drain voltage and determined by the gate voltage .
- The drain characteristic of a JFET is shown below:

JFET drain characteristic

- The advantages of JFETs over BJTs are: lower power consumption, higher input impedance, lower noise, better thermal stability, and simpler biasing circuits .
- The disadvantages of JFETs are: lower gain, lower frequency response, higher distortion, and more sensitivity to temperature and radiation .

Transfer Characteristic of FET

- A field-effect transistor (FET) is a type of transistor that uses an electric field to control the flow of current in a semiconductor.
- A FET has three terminals: source, gate, and drain. The source and drain are connected by a channel of either n-type or p-type semiconductor material, and the gate is insulated from the channel by a thin layer of oxide or other dielectric material.
- The transfer characteristic of a FET is a plot of the drain current (I_D) versus the gate-source voltage (V_GS) for a given value of the drain-source voltage (V_DS) .

- The transfer characteristic shows how the FET behaves as a voltage-controlled current source, where the gate voltage controls the amount of current flowing through the channel.
- The transfer characteristic can be derived from the drain characteristic, which is a plot of the drain current (I_D) versus the drain-source voltage (V_DS) for different values of the gate-source voltage (V_GS) .
- To derive the transfer characteristic from the drain characteristic, a line is drawn vertically on the drain characteristic to represent a constant V_DS level. The corresponding I_D and V_GS values along this line are noted and then used to plot the transfer characteristic .
- The shape of the transfer characteristic depends on the type of FET (JFET or MOSFET) and the type of channel (n-type or p-type) .
- For a JFET, the transfer characteristic is nonlinear and has a negative slope, meaning that the drain current decreases as the gate voltage becomes more negative (for n-channel) or more positive (for p-type) .
- For a MOSFET, the transfer characteristic has two regions: the cutoff region and the saturation region. In the cutoff region, the drain current is zero or very small, and the gate voltage is below a threshold value (V_T). In the saturation region, the drain current is constant and independent of V_DS, and the gate voltage is above V_T .
- A universal transfer characteristic is a transfer characteristic plotted with normalized values of I_D and V_GS, such that I_DSS = 1 and V_P = 1, where I_DSS is the maximum drain current and V_P is the pinch-off voltage for a JFET. A universal transfer characteristic can be applied to analyze or design a circuit using virtually any JFET, as long as the values of I_DSS and V_P are known for the device.

MOSFET (MOS) (Depletion and Enhancement) Type

- MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor.
- It is a type of FET that uses an electric field to control the conductivity of a channel between the source and drain terminals.
- The channel is made of a semiconductor material, such as silicon, and is covered by a thin layer of metal oxide, such as silicon dioxide.
- The gate terminal is separated from the channel by the metal oxide layer, which acts as an insulator.
- There are two major types of MOSFETs: depletion mode and enhancement mode.
- Depletion mode MOSFETs have a channel that is already formed during the manufacturing process .
- They can conduct current between the source and drain terminals even when the gate voltage is zero .
- Applying a reverse voltage to the gate reduces the channel width and

decreases the current flow .

- Applying a forward voltage to the gate increases the channel width and increases the current flow .
- Depletion mode MOSFETs can be either n-channel or p-channel, depending on the type of charge carriers in the channel.
- Enhancement mode MOSFETs have no channel during the manufacturing process .
- They do not conduct current between the source and drain terminals when the gate voltage is zero .
- Applying a forward voltage to the gate creates a channel by attracting charge carriers from the substrate .
- Applying a higher forward voltage to the gate increases the channel width and increases the current flow .
- Applying a reverse voltage to the gate does not affect the channel, as there is no channel to be reduced .
- Enhancement mode MOSFETs can also be either n-channel or p-channel, depending on the type of charge carriers in the channel.
- The symbols and characteristics of depletion and enhancement mode MOSFETs are shown below .

Symbols of depletion and enhancement mode MOSFETs

Characteristics of depletion and enhancement mode MOSFETs

- MOSFETs have many applications in electronics, such as switches, amplifiers, converters, inverters, logic gates, memory devices, sensors, and more.

Transfer Characteristic of Field Effect Transistor

- A field effect transistor (FET) is a type of transistor that uses an electric field to control the flow of current in the channel between the source and the drain terminals.
- The gate terminal is used to apply the electric field to the channel, and it is insulated from the channel by a thin layer of oxide or junction.
- The transfer characteristic of a FET is a curve that shows the relationship between the gate voltage (VGS) and the drain current (ID) for a given drain-source voltage (VDS).
- The transfer characteristic can be obtained experimentally by keeping VDS constant and measuring ID for different values of VGS.
- The shape of the transfer characteristic depends on the type and mode of operation of the FET. There are two main types of FETs: junction field effect transistors (JFETs) and metal oxide semiconductor field effect transistors (MOSFETs).
- JFETs have a channel that is doped with either n-type or p-type impurities, and a gate that is formed by a reverse-biased pn junction. JFETs can

operate in either depletion mode or enhancement mode, depending on the polarity of VGS.

- MOSFETs have a channel that is either lightly doped or undoped, and a gate that is separated from the channel by a thin layer of oxide. MOSFETs can also operate in either depletion mode or enhancement mode, depending on the polarity and magnitude of VGS.
- The transfer characteristic of a JFET in depletion mode is shown in the figure below. It can be seen that ID decreases as VGS becomes more negative (for n-channel JFET) or more positive (for p-channel JFET). This is because the reverse-biased gate junction reduces the width of the channel and increases its resistance. When VGS reaches a certain value, called the pinch-off voltage (VP), the channel is completely closed and ID becomes zero. The maximum value of ID that can flow through the JFET when VGS is zero is called the shorted gate drain current (IDSS).

Transfer characteristic of a JFET in depletion mode

- The transfer characteristic of a MOSFET in enhancement mode is shown in the figure below. It can be seen that ID increases as VGS becomes more positive (for n-channel MOSFET) or more negative (for p-channel MOSFET). This is because the applied gate voltage induces a channel of opposite polarity to the substrate under the oxide layer. When VGS reaches a certain value, called the threshold voltage (VT), the channel is formed and ID starts to flow. The slope of the transfer characteristic is called the transconductance (gm), and it measures the gain or amplification of the MOSFET.

Transfer characteristic of a MOSFET in enhancement mode

- The transfer characteristic of a FET is useful for analyzing and designing circuits that use FETs as amplifiers, switches, or sensors. By knowing the transfer characteristic, one can determine the operating point, the input and output impedances, the voltage gain, and the frequency response of the FET circuit.

Unit 4 - Operational Amplifiers

- An operational amplifier (op amp) is an analog circuit block that takes a differential voltage input and produces a single-ended voltage output.
- Op amps usually have three terminals: two high-impedance inputs and a low-impedance output port. The inverting input is denoted with a minus (-) sign, and the non-inverting input uses a positive (+) sign .
- There are four ways to classify operational amplifiers based on their input and output characteristics:
  - Voltage amplifiers take voltage in and produce a voltage at the output.
  - Current amplifiers receive a current input and produce a current output.

  - Transconductance amplifiers convert a voltage input to a current output.
  - Transresistance amplifiers convert a current input to a voltage output.
- Op amps are one of the basic building blocks of analog electronics. They can be used for various applications, such as amplification, filtering, signal conditioning, feedback, oscillation, and arithmetic operations .
- Op amps have some ideal characteristics that make them versatile and easy to use:
  - Infinite open-loop gain
  - Infinite input impedance
  - Zero output impedance
  - Infinite bandwidth
  - Zero noise
  - Zero offset voltage
- However, in reality, op amps have some limitations and non-idealities that affect their performance and require careful design and analysis:
  - Finite open-loop gain
  - Finite input impedance
  - Non-zero output impedance
  - Finite bandwidth
  - Non-zero noise
  - Non-zero offset voltage

Introduction for the notes of the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING

- An operational amplifier (op-amp) is a high-gain, direct-coupled, differential-input, linear electronic device that can perform various analog signal processing operations such as amplification, filtering, integration, differentiation, addition, subtraction, etc.
- The name operational amplifier comes from the fact that it was originally designed to perform mathematical operations using external feedback components such as resistors and capacitors.
- An op-amp has two input terminals, called the inverting (-) and the non-inverting (+) inputs, and one output terminal. The output voltage is proportional to the difference between the input voltages, multiplied by a large factor called the open-loop gain (A).
- The open-loop gain of an op-amp is very high, typically in the order of 10^5 to 10^6, and it decreases with increasing frequency. The open-loop gain also depends on the temperature, supply voltage, and manufacturing variations.
- The op-amp also has two power supply terminals, usually labeled as Vcc

and Vee, which provide the positive and negative voltages required for the operation of the op-amp. The power supply voltages can range from a few volts to tens of volts, depending on the type and specification of the op-amp.

- The op-amp is usually represented by a triangular symbol, as shown below, with the input and output terminals marked accordingly. The power supply terminals are often omitted for simplicity, but they are implied to be connected.

op-amp symbol

- The op-amp is a versatile and widely used device in electronic circuits. It can be used as a standalone component or as a part of an integrated circuit (IC). There are many types and models of op-amps available in the market, each with different characteristics and specifications. Some of the common parameters that describe the performance of an op-amp are:
  - Input impedance: The resistance offered by the op-amp to the input signal. It is usually very high, in the order of megaohms or gigaohms, to minimize the loading effect on the input source.
  - Output impedance: The resistance offered by the op-amp to the output load. It is usually very low, in the order of ohms or milliohms, to maximize the power transfer to the load.
  - Input offset voltage: The voltage difference between the input terminals when the output voltage is zero. It is ideally zero, but in practice, it is a small nonzero value due to the mismatch of the input transistors. It causes a DC error in the output voltage.
  - Input bias current: The average of the currents flowing into the input terminals. It is ideally zero, but in practice, it is a small nonzero value due to the leakage currents of the input transistors. It causes a DC error in the output voltage.
  - Input offset current: The difference between the currents flowing into the input terminals. It is ideally zero, but in practice, it is a small nonzero value due to the mismatch of the input transistors. It causes a DC error in the output voltage.
  - Common-mode rejection ratio (CMRR): The ratio of the differential gain to the common-mode gain of the op-amp. It measures the ability of the op-amp to reject the signals that are common to both input terminals. It is usually expressed in decibels (dB). It is ideally infinite, but in practice, it is a finite value that depends on the frequency and the power supply voltages.
  - Slew rate: The maximum rate of change of the output voltage with respect to time. It measures the speed of response of the op-amp to a step input. It is usually expressed in volts per microsecond (V/µs). It is limited by the internal capacitances and currents of the op-amp.
  - Bandwidth: The range of frequencies over which the op-amp can operate with a constant gain. It is usually expressed in hertz (Hz).

It is limited by the open-loop gain and the slew rate of the op-amp.

- The op-amp can be configured in different ways by using external feedback components to achieve various functions and characteristics. The feedback is the process of connecting a part of the output signal back to the input signal, either in phase (positive feedback) or out of phase (negative feedback). The feedback can affect the gain, stability, linearity, bandwidth, and noise of the op-amp circuit.

Hello, I am Sydney, your AI assistant. I can help you with your study material for the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING. Here is the content in markdown format:

Op-Amp Basics

- An operational amplifier (op-amp) is a high-gain, direct-coupled, differential-input amplifier that can amplify AC or DC signals.
- An op-amp has two input terminals: the inverting input (-) and the non-inverting input (+), and one output terminal.
- An op-amp also has two power supply terminals: the positive supply (V+) and the negative supply (V-), which provide the voltage and current for the op-amp to operate.
- The basic symbol of an op-amp is shown below:

op-amp symbol

- The ideal characteristics of an op-amp are:
  - Infinite voltage gain (A)
  - Infinite input resistance (Ri)
  - Zero output resistance (Ro)
  - Zero input offset voltage (Vos)
  - Infinite bandwidth (B)
  - Zero noise
- The actual characteristics of an op-amp are:
  - Finite voltage gain (A)
  - High input resistance (Ri)
  - Low output resistance (Ro)
  - Small input offset voltage (Vos)
  - Finite bandwidth (B)
  - Some noise
- The voltage gain (A) of an op-amp is the ratio of the output voltage (Vo) to the differential input voltage (Vd), i.e., A = Vo/Vd.
- The differential input voltage (Vd) of an op-amp is the difference between the voltages at the two input terminals, i.e., Vd = V+ - V-.
- The input offset voltage (Vos) of an op-amp is the voltage that must be applied to the input terminals to make the output voltage zero, i.e., Vos = -Vo/A.

- The bandwidth (B) of an op-amp is the range of frequencies over which the op-amp can operate with a constant gain.
- The op-amp can be used in various configurations to perform different functions, such as amplification, buffering, filtering, summing, subtracting, integrating, differentiating, etc.
- The most common configuration of an op-amp is the negative feedback configuration, where a fraction of the output voltage is fed back to the inverting input terminal through a feedback resistor (Rf).
- The negative feedback configuration reduces the gain of the op-amp, but improves its stability, linearity, bandwidth, and input and output impedances.
- The closed-loop voltage gain (Af) of an op-amp in the negative feedback configuration is the ratio of the output voltage (Vo) to the input voltage (Vi), i.e., Af = Vo/Vi.
- The closed-loop voltage gain (Af) of an op-amp in the negative feedback configuration can be calculated using the formula: Af = A/(1 + AB), where A is the open-loop voltage gain and B is the feedback factor.
- The feedback factor (B) of an op-amp in the negative feedback configuration is the ratio of the feedback voltage (Vf) to the output voltage (Vo), i.e., B = Vf/Vo.
- The feedback factor (B) of an op-amp in the negative feedback configuration can be calculated using the formula: B = R1/(R1 + Rf), where R1 is the input resistor and Rf is the feedback resistor.
- The negative feedback configuration of an op-amp can be classified into two types: the inverting amplifier and the non-inverting amplifier.
- The inverting amplifier is a negative feedback configuration where the input voltage (Vi) is applied to the inverting input terminal (-) and the output voltage (Vo) is inverted with respect to the input voltage, i.e., Vo = -AfVi.
- The non-inverting amplifier is a negative feedback configuration where the input voltage (Vi) is applied to the non-inverting input terminal (+) and the output voltage (Vo) is in phase with the input voltage, i.e., Vo = AfVi.
- The circuit diagrams and the formulas for the closed-loop voltage gain (Af) of the inverting amplifier and the non-inverting amplifier are shown below:

inverting amplifier

Af = -Rf/R1

non-inverting amplifier(https://

Practical Op-Amp Circuits

Operational amplifiers (op-amps) are versatile and widely used electronic devices that can perform various functions such as amplification, filtering, integration, differentiation, etc. In this note, we will discuss some of the common and useful op-amp circuits that can be used for various applications.

1. Voltage Follower

The voltage follower is the simplest op-amp circuit, as it does not require any external components. It is also called a buffer, as it isolates the input from the output and prevents loading effects. The voltage follower has a unity gain, which means that the output voltage is equal to the input voltage. The circuit diagram and the input-output characteristics are shown below.

Voltage follower circuit diagram

Voltage follower input-output characteristics

The voltage follower can be used to:

- Provide impedance matching between different stages of a circuit
- Drive low-impedance loads such as speakers or LEDs
- Extend the bandwidth of a circuit by reducing the capacitive loading
- Protect sensitive inputs from noise or interference

2. Inverting Op-Amp

The inverting op-amp is a basic circuit that can perform voltage amplification with a negative gain. The output voltage is proportional to the input voltage, but with an opposite polarity. The gain of the circuit is determined by the ratio of the feedback resistor R2 to the input resistor R1. The circuit diagram and the input-output characteristics are shown below.

Inverting op-amp circuit diagram

Inverting op-amp input-output characteristics

The inverting op-amp can be used to:

- Amplify signals with a desired gain and polarity
- Perform mathematical operations such as subtraction and scaling
- Implement active filters such as low-pass, high-pass, band-pass, etc.
- Generate oscillations and waveforms

3. Non-inverting Op-Amp

The non-inverting op-amp is another basic circuit that can perform voltage amplification with a positive gain. The output voltage is proportional to the input voltage, but with the same polarity. The gain of the circuit is determined by the ratio of the feedback resistor R2 to the input resistor R1, plus one. The circuit diagram and the input-output characteristics are shown below.

Non-inverting op-amp circuit diagram

Inverting Amplifier

- An inverting amplifier is a type of operational amplifier circuit that produces an output signal that is 180 degrees out of phase with the input signal .
- This means that if the input signal is positive, then the output signal will be negative and vice versa .
- An inverting amplifier can be used to amplify or invert a signal, depending on the values of the resistors in the circuit .
- An inverting amplifier can also be used for signal conditioning or mathematical operations, such as subtraction, integration, differentiation, etc .

Circuit Diagram

- The basic circuit diagram of an inverting amplifier is shown below:

Inverting amplifier circuit diagram

- The circuit consists of an operational amplifier (op-amp), an input resistor (Ri), and a feedback resistor (Rf).
- The input signal (Vin) is applied to the inverting input terminal (-) of the op-amp, while the non-inverting input terminal (+) is connected to the ground.
- The output signal (Vout) is taken from the output terminal of the op-amp.

Working Principle

- The working principle of an inverting amplifier is based on the negative feedback model, which means that a fraction of the output signal is fed back to the input through the feedback resistor (Rf) .
- The op-amp tries to maintain the same voltage at both of its input terminals, which is called the virtual ground principle .
- This means that the voltage at the inverting input terminal (-) is equal to the voltage at the non-inverting input terminal (+), which is zero in this case .
- Therefore, the voltage across the input resistor (Ri) is equal to the input signal (Vin), and the current through the input resistor (Ii) is given by:

Ii = Vin / Ri

- Since the op-amp has a very high input impedance, the current through the inverting input terminal (-) is negligible, and the same current (Ii) flows through the feedback resistor (Rf) .
- The voltage across the feedback resistor (Rf) is equal to the output signal (Vout), and the current through the feedback resistor (Ii) is given by:

Ii = Vout / Rf

- Equating the two expressions for the current (Ii), we get:

Vin / Ri = Vout / Rf

- Rearranging the equation, we get the voltage gain (Av) of the inverting amplifier, which is given by:

Av = Vout / Vin = - Rf / Ri

- The negative sign indicates that the output signal is inverted with respect to the input signal .
- The magnitude of the voltage gain depends on the ratio of the feedback resistor (Rf) to the input resistor (Ri) .
- If Rf > Ri, the voltage gain is greater than 1, and the circuit acts as an inverting amplifier .
- If Rf < Ri, the voltage gain is less than 1, and the circuit acts as an inverting attenuator .
- If Rf = Ri, the voltage gain is equal to 1, and the circuit acts as an inverting buffer .

Applications

- Some of the applications of an inverting amplifier are:
  - Signal inversion: An inverting amplifier can be used to invert the polarity of a signal, such as converting a positive signal to a negative signal or vice versa .
  - Signal amplification: An inverting amplifier can be used to amplify a signal by choosing a suitable value of the feedback resistor (Rf) that is larger than the input resistor (Ri) [

Non-inverting Amplifier

- A non-inverting amplifier is an op-amp circuit configuration that produces an amplified output signal and this output signal of the non-inverting op-amp is in-phase with the applied input signal .
- In other words, a non-inverting amplifier behaves like a voltage follower circuit.
- The basic circuit diagram of a non-inverting amplifier is shown below:

Non-inverting amplifier circuit diagram

- The input voltage signal, (VIN) is applied directly to the non-inverting (+) input terminal which means that the output gain of the amplifier becomes “Positive” in value in contrast to the “Inverting Amplifier” circuit we saw in the previous tutorial whose output gain is negative in value.
- The feedback resistor, RF and the input resistor, RIN form a potential divider network across the amplifier and the voltage gain of a non-inverting amplifier can be calculated as :

Non-inverting amplifier voltage gain formula

- The voltage gain of a non-inverting amplifier is always greater than one .
- The input impedance of a non-inverting amplifier is very high, as the input signal is applied to the non-inverting input terminal of the op-amp, which has a very high input impedance .
- The output impedance of a non-inverting amplifier is very low, as the output signal is taken from the output terminal of the op-amp, which has a very low output impedance .
- The non-inverting amplifier has some advantages over the inverting amplifier, such as no phase inversion, higher input impedance, lower output impedance, and higher bandwidth.
- The non-inverting amplifier has some applications, such as buffer amplifier, impedance converter, voltage follower, summing amplifier, and differential amplifier.

Unit Follower

- A unit follower is a type of operational amplifier (op-amp) circuit that has a voltage gain of 1.
- It is also called a voltage follower, an op-amp buffer, or a unity-gain amplifier.
- It is used to isolate circuits from each other, to prevent loading effects, and to provide impedance matching.
- It consists of an op-amp with a direct feedback connection from the output to the inverting (-) input.
- The non-inverting (+) input is connected to the input voltage source.
- The output voltage is equal to the input voltage, hence the name voltage follower.
- The input impedance of the unit follower is very high, ideally infinite, so it does not draw any current from the input source.
- The output impedance of the unit follower is very low, ideally zero, so it can drive any load without voltage drop.
- The unit follower has a phase shift of zero degrees, meaning it does not invert the input signal.
- The unit follower has a bandwidth of almost the entire frequency range of the op-amp, since it does not have any external resistors or capacitors that affect the frequency response.
- The unit follower can be used for various applications, such as signal buffering, impedance transformation, level shifting, and voltage reference.

The following diagram shows the circuit of a unit follower using an op-amp:

Unit follower circuit

The following equations describe the operation of the unit follower:

Vout = Vin

Vout - V- = 0

V+ = V-

V+ = Vin

Vout = Vin

The following table summarizes the characteristics of the unit follower:

Parameter	Value
Voltage gain	1
Input impedance	Very high
Output impedance	Very low
Phase shift	0 degrees
Bandwidth	Almost the entire op-amp frequency range

Summing Amplifier

- A summing amplifier is an op amp circuit that can combine numbers of input signals to a single output that is the weighted sum of the applied inputs .
- The summing amplifier is one variation of inverting amplifier. In inverting amplifier there is only one voltage signal applied to the inverting input as shown below:

Inverting amplifier

- In summing amplifier, there are two or more voltage signals applied to the inverting input through different resistors as shown below:

Summing amplifier

- The output voltage of the summing amplifier is given by the formula :

Output voltage formula

- The summing amplifier can be used to perform arithmetic operations such as addition, subtraction, scaling, and averaging of the input signals .
- The summing amplifier can also be used to implement digital-to-analog converters, audio mixers, and weighted summing networks .

Integrator

Definition

- An integrator is an electronic circuit that performs the mathematical operation of integration with respect to time.
- It is based on an operational amplifier (op-amp) with a capacitor and a resistor in the feedback loop.
- The output voltage of the integrator is proportional to the input voltage integrated over time.

Circuit Diagram

- The basic circuit diagram of an op-amp integrator is shown below:

Op-amp integrator circuit diagram

- The input voltage is applied to the inverting terminal of the op-amp through a resistor R.
- The output voltage is fed back to the inverting terminal through a capacitor C.
- The non-inverting terminal is grounded.

Working Principle

- The working principle of the op-amp integrator can be explained using the virtual ground concept and the capacitor charge equation.
- The virtual ground concept states that the voltage at the inverting terminal of the op-amp is equal to the voltage at the non-inverting terminal, which is zero in this case.
- Therefore, the voltage across the resistor R is equal to the input voltage Vin.
- The current flowing through the resistor R is given by:

I = Vin / R

- This current also flows through the capacitor C, since the op-amp has a very high input impedance and draws negligible current.
- The capacitor charge equation states that the voltage across the capacitor C is equal to the charge on the capacitor divided by its capacitance, which is given by:

VC = Q / C

- The charge on the capacitor Q is equal to the integral of the current over time, which is given by:

Q = I dt

- Substituting the values of I and VC, we get:

VC = (1 / C)   Vin / R dt

- The output voltage Vout is equal to the negative of the voltage across the capacitor C, since the op-amp is in inverting configuration. Therefore, we get:

Vout = - VC

- Substituting the value of VC, we get:

Vout = - (1 / C)   Vin / R dt

- This equation shows that the output voltage of the op-amp integrator is proportional to the integral of the input voltage over time, with a constant of proportionality of -1 / RC.

Applications

- The op-amp integrator can be used for various applications, such as:
  - Signal processing: The op-amp integrator can be used to perform mathematical operations on signals, such as differentiation, integration, filtering, etc.
  - Waveform generation: The op-amp integrator can be used to generate different types of waveforms, such as triangular, sawtooth, square, etc., by applying different types of input signals, such as sinusoidal, square, etc.
  - Analog computation: The op-amp integrator can be used to perform analog computation, such as solving differential equations, calculating areas, etc., by applying the appropriate input signals and feedback components.

Differentiator

A differentiator is a circuit that performs the mathematical operation of differentiation, that is, it produces an output voltage that is proportional to the rate of change of the input voltage. A differentiator can be constructed using an operational amplifier (op-amp) with a capacitor and a resistor in the feedback loop. The differentiator is one of the applications of op-amps in analog signal processing.

Circuit Diagram

The circuit diagram of a differentiator using an op-amp is shown below:

Differentiator circuit diagram

The input voltage is applied to the inverting terminal of the op-amp through a capacitor C, while the non-inverting terminal is grounded. A resistor R is connected between the output and the inverting terminal to provide negative feedback. The output voltage is taken across the resistor R.

Working Principle

The working principle of the differentiator can be explained using the capacitor's current to voltage relationship. The current through a capacitor is given by:

`i = C dv/dt`

where i is the current, C is the capacitance, v is the voltage across the capacitor, and t is the time.

The voltage across the capacitor is equal to the input voltage, since the inverting terminal of the op-amp is virtually grounded. Therefore, the current through the capacitor is:

`i = C dV_in/dt`

The current through the capacitor is also equal to the current through the resistor, since the op-amp has a very high input impedance and a very low output impedance. Therefore, the voltage across the resistor is:

`V_out = -iR = -RC dV_in/dt`

The negative sign indicates that the output voltage is 180 degrees out of phase with the input voltage. The output voltage is proportional to the rate of change of the input voltage, which is the definition of differentiation.

Frequency Response

The frequency response of the differentiator can be obtained by applying a sinusoidal input voltage of the form:

`V_in = V_m sin( t)`

where V_m is the peak voltage, is the angular frequency, and t is the time.

The output voltage is then:

`V_out = -RC dV_in/dt = -RC V_m  cos( t)`

The magnitude of the output voltage is:

`|V_out| = RC V_m`

The phase difference between the input and output voltages is:

` = tan^(-1)(-1/RC )`

The frequency response of the differentiator is shown below:

Differentiator frequency response

The frequency response shows that the differentiator acts as a high-pass filter, that is, it passes high-frequency signals and attenuates low-frequency signals. The gain of the differentiator increases linearly with frequency, which means that the output voltage can become very large for high-frequency signals. This can cause noise and distortion in the output signal. To avoid this, a resistor R1 can be added in series with the capacitor C, as shown below:

Differentiator with R1

The resistor R1 limits the gain of the differentiator at high frequencies and improves the stability of the circuit. The output voltage of the modified differentiator is:

```
V_out = -R/(R1 + 1/ C) dV_in/dt
```

The frequency response of the modified differentiator is shown below:

Modified differentiator frequency response

The frequency response shows that the modified differentiator has a constant gain at low frequencies and a decreasing gain at high frequencies. The cutoff frequency of the modified differentiator is:

```
f_c = 1/(2 R1C)
```

The cutoff frequency is the frequency at which the gain of the differentiator drops by 3 dB from its maximum value. The cutoff frequency determines the range of frequencies over which the differentiator can perform accurate differentiation.

Applications

The differentiator can be used in various applications, such as:

- Waveform generation: The differentiator can be used to generate square, triangular, and sawtooth waveforms from sinusoidal inputs.
- Edge detection: The differentiator can be used to detect the

Differential and Common-Mode Operation of Op-Amp

- An op-amp is a differential amplifier that can amplify the difference between two input signals.
- The input signals can be classified into two modes: common-mode and differential-mode.
- Common-mode signals are the signals that are the same for both inputs, while differential-mode signals are the signals that are different for both inputs.

- For example, if the input signals are V1 and V2, then the common-mode signal is (V1 + V2) / 2 and the differential-mode signal is (V1 - V2) / 2.
- The op-amp ideally amplifies only the differential-mode signal and rejects the common-mode signal. This is because the op-amp has a high common-mode rejection ratio (CMRR), which is the ratio of the differential-mode gain to the common-mode gain.
- The common-mode gain is the gain of the op-amp when both inputs are connected to the same signal, while the differential-mode gain is the gain of the op-amp when the inputs are connected to different signals.
- The common-mode gain is usually very small compared to the differential-mode gain, so the CMRR is very large, typically in the order of 10^5 or more.
- The common-mode signal can cause interference or noise in the op-amp output, especially if the input signals are not well balanced or matched. Therefore, it is desirable to minimize the common-mode signal and maximize the differential-mode signal for better performance and accuracy of the op-amp.
- The common-mode and differential-mode signals can be represented by the following equations:
  - Vcm = (V1 + V2) / 2
  - Vd = (V1 - V2) / 2
  - Vout = Ad * Vd + Ac * Vcm
  - where Ad is the differential-mode gain, Ac is the common-mode gain, Vcm is the common-mode signal, Vd is the differential-mode signal, and Vout is the output signal of the op-amp.
- The common-mode and differential-mode signals can also be represented by the following diagram:

  Diagram of common-mode and differential-mode signals

  - Figure 1: Diagram of common-mode and differential-mode signals

Comparators for the notes of the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING

- A comparator is a circuit that uses an operational amplifier to compare two voltages and output a high or low signal depending on which voltage is larger .
- A comparator can be used for various applications, such as polarity identification, analog to digital conversion, switch driving, waveform generation,

and pulse-edge detection .

- A comparator can be configured by using an operational amplifier in its open-loop state, that is, without any feedback resistor. This allows the op-amp to have a very high gain and a very fast response time.
- A comparator has two inputs, called the inverting input (-) and the non-inverting input (+), and one output. The output voltage is determined by the following rules:
  - If the non-inverting input voltage is greater than the inverting input voltage, the output voltage is equal to the positive supply voltage (+Vcc).
  - If the non-inverting input voltage is less than the inverting input voltage, the output voltage is equal to the negative supply voltage (-Vcc).
  - If the non-inverting input voltage is equal to the inverting input voltage, the output voltage is undefined and may oscillate between +Vcc and -Vcc.
- A comparator can be classified into different types based on the number and values of the reference voltages used to compare the input voltage. Some common types of comparators are:
  - Zero-crossing detector: A comparator that uses zero volts as the reference voltage and detects when the input voltage crosses the zero level.
  - Level detector: A comparator that uses a fixed reference voltage and detects when the input voltage reaches or exceeds that level.
  - Window detector: A comparator that uses two reference voltages and detects when the input voltage is within or outside a certain range or window of values.
  - Hysteresis comparator: A comparator that uses positive feedback to introduce a small difference between the reference voltages and prevent output oscillations due to noise or fluctuations in the input voltage.

Unit 5 - Digital Electronics

- Digital electronics is the branch of electronics that deals with the representation and manipulation of data in digital form.
- It involves the use of devices such as transistors, diodes, and microcontrollers to process and transmit digital signals.
- Digital signals are binary, meaning they can only have two values: 0 or 1, also called low or high, false or true, off or on.
- Digital electronics are electronic fields that include the area of digital signals and engineering elements that yields high productivity.
- It is comprised of multiple logic gates packed as integrated circuits.
- Logic gates are the building blocks of all digital electronic circuits.
- They perform basic logical operations such as AND, OR, and NOT on the

input signals and produce an output signal.

- The output of a logic gate depends on the truth table of the gate, which shows the output for every possible combination of inputs.
- Digital electronics has many advantages over analog electronics, such as:
  - Higher speed and accuracy.
  - Lower power consumption and cost.
  - Easier storage and transmission of data.
  - Greater noise immunity and error correction.
- Digital electronics has many applications in various fields, such as:
  - Computers and microprocessors.
  - Communication and networking.
  - Digital signal processing and image processing.
  - Robotics and automation.
  - Biomedical and healthcare.
- Digital electronics is a field of electronics involving the study of digital signals and the engineering of devices that use or produce them.
- It requires the knowledge of Boolean algebra, binary arithmetic, logic gates, combinational and sequential circuits, and memory devices.
- It also requires the skills of designing, testing, and debugging digital circuits using software tools such as LabVIEW.

Number System and Representation for the Notes of the Unit 5 - Digital Electronics in the Subject of Fundamentals of Electronics Engineering

- A number system is a way of representing information using symbols or digits.
- Different number systems have different bases or radices, which are the total number of symbols or digits used in the system.
- The most common number systems in digital electronics are decimal, binary, octal, and hexadecimal.
- Decimal number system has a base of 10 and uses 10 symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
- Binary number system has a base of 2 and uses 2 symbols: 0 and 1. Each symbol is also called a bit.
- Octal number system has a base of 8 and uses 8 symbols: 0, 1, 2, 3, 4, 5, 6, and 7.
- Hexadecimal number system has a base of 16 and uses 16 symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F.
- The value of a number in any system depends on its position and the base of the system. For example, the decimal number 2843 can be written as:

2843 = 2 x 10^3 + 8 x 10^2 + 4 x 10^1 + 3 x 10^0

- Similarly, the binary number 1010 can be written as:

1010 = 1 x 2^3 + 0 x 2^2 + 1 x 2^1 + 0 x 2^0

- To convert a number from one system to another, we can use different methods such as division, multiplication, substitution, or table lookup.
- For example, to convert the decimal number 2843 to binary, we can use the division method as follows:

2843 / 2 = 1421 remainder 1

1421 / 2 = 710 remainder 1

710 / 2 = 355 remainder 0

355 / 2 = 177 remainder 1

177 / 2 = 88 remainder 1

88 / 2 = 44 remainder 0

44 / 2 = 22 remainder 0

22 / 2 = 11 remainder 0

11 / 2 = 5 remainder 1

5 / 2 = 2 remainder 1

2 / 2 = 1 remainder 0

1 / 2 = 0 remainder 1

- The binary equivalent of 2843 is the remainders in reverse order: 101100100011
- To convert the binary number 1010 to decimal, we can use the multiplication method as follows:

1010 = 1 x 2^3 + 0 x 2^2 + 1 x 2^1 + 0 x 2^0

= 1 x 8 + 0 x 4 + 1 x 2 + 0 x 1

= 8 + 0 + 2 + 0

= 10

- The decimal equivalent of 1010 is 10.
- To convert the octal number 5347 to decimal, we can use the multiplication method as follows:

5347 = 5 x 8^3 + 3 x 8^2 + 4 x 8^1 + 7 x 8^0

= 5 x 512 + 3 x 64 + 4 x 8 + 7 x 1

= 2560 + 192 + 32 + 7

= 2791

- The decimal equivalent of 5347 is 2791.
- To convert the hexadecimal number 2A3F to decimal, we can use the multiplication method as follows:

2A3F = 2 x 16^3 + A x 16^2 + 3 x 16^1 + F x 16^0

= 2 x 4096 + 10 x 256 + 3 x 16 + 15 x 1

= 8192 + 2560 + 48 + 15

= 10815

- The decimal equivalent of 2A3F is 10815.
- To convert a number from decimal to octal or hexadecimal

Binary arithmetic

Binary arithmetic is a set of rules for performing arithmetic operations on numbers represented in binary form. Binary numbers have only two digits: 0 and 1. Binary arithmetic operations include binary addition, binary subtraction, binary multiplication, and binary division. These operations are simpler than decimal arithmetic operations because they involve fewer cases and rules.

Binary addition

Binary addition is the simplest and most basic operation of binary arithmetic. It is used to add two binary numbers and produce a binary sum. The rules of binary addition are based on the truth table shown below:

A	B	Carry	Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

The carry bit is generated when both the bits are 1 and is added to the next pair of bits. The sum bit is the result of the exclusive OR (XOR) operation on the two bits. Binary addition is performed from right to left, starting from the least significant bit (LSB).

For example, to add 1011 and 1101, we follow these steps:

Step	A	B	Carry	Sum
1	1011	1101	0	
2	1011	1101	0	0
3	1011	1101	1	10

Step	A	B	Carry	Sum
4	1011	1101	1	000
5	1011	1101	1	1000

The final sum is 10000.

Binary subtraction

Binary subtraction is another basic operation of binary arithmetic. It is used to subtract one binary number from another and produce a binary difference. The rules of binary subtraction are based on the truth table shown below:

A	B	Borrow	Difference
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

The borrow bit is generated when the first bit is 0 and the second bit is 1 and is subtracted from the next pair of bits. The difference bit is the result of the XOR operation on the two bits. Binary subtraction is performed from right to left, starting from the LSB.

For example, to subtract 1010 from 1100, we follow these steps:

Step	A	B	Borrow	Difference
1	1100	1010	0	
2	1100	1010	0	0
3	1100	1010	0	10
4	1100	1010	1	010
5	1100	1010	0	0010

The final difference is 10.

Binary multiplication

Binary multiplication is a more complex operation of binary arithmetic. It is used to multiply two binary numbers and produce a binary product. The rules of binary multiplication are based on the truth table shown below:

A	B	Product
0	0	0
0	1	0
1	0	0
1	1	1

The product bit is the result of the AND operation on the two bits. Binary multiplication is performed by multiplying each bit of the first number by each bit of the second number and adding the partial products. The partial products are shifted left by one bit for each position of the multiplier.

For example, to multiply 101 and 110, we follow these steps:

Step	A	B	Partial product
1			

Introduction of Basic and Universal Gates

- Logic gates are the fundamental building blocks of digital electronics. They perform basic logical operations on binary inputs and produce a single binary output.
- There are three basic logic gates: AND, OR, and NOT. They have the following symbols and truth tables:

AND gate	OR gate	NOT gate
AND gate symbol	OR gate symbol	NOT gate symbol
AND gate truth table	OR gate truth table	NOT gate truth table

- The AND gate produces a 1 output only when both inputs are 1. The OR gate produces a 1 output when either or both inputs are 1. The NOT gate produces the inverse of the input.
- There are two universal gates: NAND and NOR. They are called universal because they can be used to implement any other logic gate or Boolean function .
- The NAND gate is the negation of the AND gate, and the NOR gate is the negation of the OR gate. They have the following symbols and truth tables:

NAND gate	NOR gate
NAND gate symbol	NOR gate symbol
NAND gate truth table	NOR gate truth table

- The NAND gate produces a 0 output only when both inputs are 1. The NOR gate produces a 0 output when either or both inputs are 1.
- To implement any other logic gate using universal gates, we can use the following rules:

Logic gate	NAND implementation	NOR implementation
AND	AND using NAND	AND using NOR
OR	OR using NAND	OR using NOR
NOT	NOT using NAND	NOT using NOR
XOR	XOR using NAND	XOR using NOR
XNOR	XNOR using NAND	XNOR using NOR

- Logic gates can be integrated into a single IC (integrated circuit) to design various processors and controllers. The IC number of a logic gate indicates its type, number of inputs, and logic family. For example, the IC number 7400 is a NAND gate with four inputs and belongs to the TTL (trans

Using Boolean Algebra Simplification of Boolean Function

- Boolean algebra is a branch of mathematics that deals with binary values, such as 0 and 1, and logical operations, such as AND, OR, NOT, XOR, etc.
- Boolean algebra can be used to represent and manipulate the logic functions of digital circuits, such as gates, flip-flops, multiplexers, etc.
- Simplification of Boolean functions is the process of finding an equivalent expression that uses fewer terms and/or operations, which leads to simpler and cheaper implementations of the circuits.
- Simplification of Boolean functions can be done by using the theorems and rules of Boolean algebra, such as identity, commutativity, associativity, distributivity, complementarity, De Morgan's laws, etc.
- Simplification of Boolean functions can also be done by using methods such as Karnaugh maps, Quine-McCluskey algorithm, etc.

Example of Simplification of Boolean Function

- Consider the following Boolean function:

  F = A.B + A.B + B.C

- The logic diagram for this function is:

  Logic diagram for F = A.B + A.B + B.C

- To simplify this function using Boolean algebra, we can apply the following steps:

- Step 1: Use the idempotent law (X + X = X) to eliminate the repeated term A.B:

  F = A.B + B.C

- Step 2: Use the distributive law (X + Y.Z = (X + Y).(X + Z)) to factor out A:

  F = A.(B + C) + B.C

- Step 3: Use the distributive law again to expand the term A.(B + C):

  F = A.B + A.C + B.C

- Step 4: Use the absorption law (X + X.Y = X) to eliminate the redundant term A.B:

  F = A + B.C

- The simplified logic diagram for this function is:

  Logic diagram for F = A + B.C

- The simplified function uses fewer gates and inputs, which reduces the cost and complexity of the circuit.

K Map Minimization upto 6 Variables

- Karnaugh map or K-map is a map of a function used in a technique used for minimization or simplification of a Boolean expression.
- It results in less number of logic gates and inputs to be used during the fabrication.
- K-map is a graphical representation of a truth table, where each cell corresponds to a minterm or a maxterm of the function.
- The cells are arranged in such a way that adjacent cells differ by only one bit position.
- The cells are also labeled with a binary address code that indicates the values of the variables for that cell.
- The main steps of K-map minimization are:
  - Identify all the essential prime implicants of the function, which are the largest groups of 1s (for SOP) or 0s (for POS) that cover at least one minterm or maxterm that is not covered by any other group.
  - Select the minimum number of essential prime implicants that cover all the minterms or maxterms of the function.
  - Write the simplified expression by taking the OR (for SOP) or AND (for POS) of the essential prime implicants.
- K-map of 2 to 4 variables is very easy. However, 5 and 6 variable K-map is a little bit complex.

- For 5 variable K-map, there are 2^5 = 32 cells, which can be arranged in a 4x8 or 8x4 matrix.
- The fifth variable is represented by two submaps, one for its value being 0 and the other for its value being 1.
- The cells in each submap are labeled with a 4-bit address code that follows the Gray code sequence.
- The cells in the same position in both submaps are considered adjacent and can be grouped together.
- For 6 variable K-map, there are 2^6 = 64 cells, which can be arranged in a 8x8 or 16x4 matrix.
- The sixth variable is represented by four submaps, one for each combination of its value and the value of the fifth variable.
- The cells in each submap are labeled with a 4-bit address code that follows the Gray code sequence.
- The cells in the same position in all four submaps are considered adjacent and can be grouped together.
- The cells in the same position in two adjacent submaps are also considered adjacent and can be grouped together.
- The cells in the corners of each submap are also considered adjacent and can be grouped together.
- The cells in the corners of the whole map are also considered adjacent and can be grouped together.
- The following diagrams show the 5 and 6 variable K-maps with the cell labels and the adjacency rules :

5 variable K-map

6 variable K-map

- The following example shows how to simplify a 6 variable function using K-map:

6 variable K-map example

- The simplified expression is:

F = A'B'C'D' + A'BCD + AB'C'D + ABCD' + EF

Unit 6 - Fundamentals of Communication Engineering

- Communication engineering is a field that focuses on supporting systems that transfer information from one place to another.
- Communication engineering involves the design, installation and maintenance of telecommunication equipment.
- Communication engineering covers the following basic concepts:

- Signals, spectra, and bandwidth: Signals are the physical representation of information, spectra are the frequency components of signals, and bandwidth is the range of frequencies that a signal occupies.
- Attenuation, distortion, and noise: Attenuation is the loss of signal power due to distance or medium, distortion is the alteration of signal shape or quality due to non-linearities or interference, and noise is the unwanted random fluctuations in signal amplitude or frequency.
- Filtering, equalizing, and companding: Filtering is the process of removing unwanted frequency components from a signal, equalizing is the process of compensating for the distortion caused by the channel or medium, and companding is the process of compressing and expanding the dynamic range of a signal to reduce the effects of noise.
- Modulation and multiplexing: Modulation is the process of changing the characteristics of a carrier signal (such as amplitude, frequency, or phase) according to the information signal, and multiplexing is the process of combining multiple signals into one signal for transmission over a common channel.
- Information measurement, source and channel coding, channel capacity and Shannon theorems: Information measurement is the quantification of the amount of information in a signal or source, source coding is the process of reducing the redundancy or inefficiency of a signal or source, channel coding is the process of adding redundancy or error correction to a signal or source, channel capacity is the maximum rate of information that can be transmitted over a channel, and Shannon theorems are the fundamental limits and principles of communication theory.
- Baseband data transmission, digital modulation and spread-spectrum systems: Baseband data transmission is the transmission of digital signals without modulation, digital modulation is the modulation of digital signals using discrete values of carrier signal characteristics, and spread-spectrum systems are the systems that use a wide bandwidth to transmit signals with low power and high security.
- Local area networks and architecture and performance: Local area networks are the networks that connect devices within a limited geographical area, such as a building or campus, and their architecture and performance depend on the topology, medium, protocol, and traffic characteristics of the network.
- Layered network architecture, protocols, switching techniques, TCP/IP, traffic engineering and basic capacity analysis: Layered network architecture is the division of network functions into hierarchical levels or layers, protocols are the rules and formats for communication between network devices, switching techniques are the methods of transferring data between network devices, TCP/IP is the most widely used network protocol suite, traffic engineering is the optimization of network performance and resource utilization,

and basic capacity analysis is the estimation of network throughput and delay.

Basics of signal representation and analysis

- A signal is a physical quantity that varies with time, space or any other independent variable. It can represent information such as sound, image, temperature, etc.
- Signal representation is the process of describing a signal using mathematical functions or symbols. Signal representation can be done in different domains, such as time, frequency, space, etc.

- Signal analysis is the process of extracting useful information from a signal, such as its amplitude, frequency, phase, energy, spectrum, etc. Signal analysis can be done using various techniques, such as Fourier analysis, Laplace transform, Z-transform, etc.

- Time domain representation is the most common way of representing a signal as a function of time. Time domain representation shows the variation of the signal amplitude with respect to time. Time domain representation is useful for understanding the temporal characteristics of a signal, such as its duration, periodicity, etc.

- Frequency domain representation is another way of representing a signal as a function of frequency. Frequency domain representation shows the variation of the signal amplitude with respect to frequency. Frequency domain representation is useful for understanding the spectral characteristics of a signal, such as its bandwidth, harmonics, etc.

- To convert a signal from time domain to frequency domain, or vice versa, we can use mathematical tools such as Fourier transform, Laplace transform, Z-transform, etc. These tools decompose a signal into its frequency components, or synthesize a signal from its frequency components.

- Some signals can be represented in both time and frequency domains, such as sinusoidal signals, exponential signals, etc. Some signals can only be represented in one domain, such as impulse signals, step signals, etc.

- Signal representation and analysis are important for communication engineering, as they help us to design, implement and evaluate various communication systems and components, such as modulators, demodulators, filters, amplifiers, etc.

Electromagnetic spectrum

- The electromagnetic spectrum is the range of all types of electromagnetic radiation, which are forms of energy that travel and spread out as they go .
- Electromagnetic radiation can be described by its frequency, wavelength, or photon energy, which are related by the equation: c = f , where c is the speed of light, f is the frequency, and is the wavelength .
- The electromagnetic spectrum covers electromagnetic waves with frequencies ranging from below one hertz to above 10^{25} hertz, corresponding to wavelengths from thousands of kilometers down to a fraction of the size of an atomic nucleus.
- The electromagnetic spectrum is divided into different regions based on the properties and applications of the electromagnetic waves. Some of the common regions are:
  - Radio waves: These have the lowest frequencies and longest wavelengths, and are used for communication, broadcasting, radar, and navigation .
  - Microwaves: These have higher frequencies and shorter wavelengths than radio waves, and are used for cooking, heating, wireless networks, and satellite communication .
  - Infrared: These have frequencies between microwaves and visible light, and are emitted by warm objects. They are used for thermal imaging, remote sensing, night vision, and security systems .
  - Visible light: These are the electromagnetic waves that humans can see with their eyes, and have frequencies between infrared and ultraviolet. They are used for vision, photography, illumination, and optical communication .
  - Ultraviolet: These have higher frequencies and shorter wavelengths than visible light, and are emitted by hot objects and stars. They are used for sterilization, disinfection, fluorescence, and sun tanning, but can also cause skin damage and cancer .
  - X-rays: These have even higher frequencies and shorter wavelengths than ultraviolet, and are produced by accelerating electrons or nuclear transitions. They are used for medical imaging, security scanning, and crystallography, but can also be harmful to living tissues .
  - Gamma rays: These have the highest frequencies and shortest wavelengths of the electromagnetic spectrum, and are generated by radioactive decay, nuclear reactions, and cosmic events. They are used for nuclear medicine, radiation therapy, and astronomy, but can also be lethal to living organisms .
- The electromagnetic spectrum is continuous, meaning that there are no gaps or boundaries between the different regions. However, the regions are defined by convention and convenience, and may vary depending on the context and source .

Elements of a Communication System

A communication system is a system that enables the exchange of information between two or more points. It consists of the following basic elements :

- **Information source**: This is the origin of the information that needs to be communicated. It can be a person, a device, or a phenomenon that generates a message. Examples of information sources are a microphone, a camera, a keyboard, a sensor, etc.
- **Input transducer**: This is a device that converts the message from the information source into a suitable form for transmission. It can be an analog or a digital device, depending on the nature of the message and the channel. Examples of input transducers are a microphone, a camera, an analog-to-digital converter, etc.
- **Transmitter**: This is a device that modulates the message signal with a carrier signal and amplifies it to a suitable level for transmission. It can be an analog or a digital device, depending on the type of modulation and the channel. Examples of transmitters are a radio transmitter, a telephone transmitter, a satellite transmitter, etc.
- **Channel**: This is the medium through which the message signal travels from the transmitter to the receiver. It can be a wired or a wireless medium, depending on the type of transmission and the distance. Examples of channels are a copper wire, a coaxial cable, an optical fiber, air, space, etc.
- **Receiver**: This is a device that demodulates the message signal from the carrier signal and amplifies it to a suitable level for output. It can be an analog or a digital device, depending on the type of modulation and the channel. Examples of receivers are a radio receiver, a telephone receiver, a satellite receiver, etc.
- **Output transducer**: This is a device that converts the message signal from the receiver into a suitable form for the destination. It can be an analog or a digital device, depending on the nature of the message and the destination. Examples of output transducers are a speaker, a monitor, a digital-to-analog converter, etc.
- **Destination**: This is the final point of the communication system where the information is delivered. It can be a person, a device, or a phenomenon that receives the message. Examples of destinations are a speaker, a monitor, a printer, a sensor, etc.

The communication system can be classified into different types based on the following criteria:

- **Analog or digital**: This depends on whether the message signal and the carrier signal are continuous or discrete in time and amplitude. Analog communication systems use continuous signals, while digital communication systems use discrete signals.
- **Simplex, half-duplex, or full-duplex**: This depends on whether the

communication is one-way, two-way with alternation, or two-way simultaneously. Simplex communication systems allow only one direction of communication, while half-duplex communication systems allow both directions of communication but not at the same time. Full-duplex communication systems allow both directions of communication at the same time.
- **Point-to-point or broadcast**: This depends on whether the communication is between two specific points or between one point and many points. Point-to-point communication systems connect only two points, while broadcast communication systems connect one point to many points.
- **Wired or wireless**: This depends on whether the channel is a physical medium or a free space. Wired communication systems use cables or wires as the channel, while wireless communication systems use electromagnetic waves as the channel.

The communication system can be represented by a block diagram as shown below :

Block diagram of a communication system

The communication system is influenced by various factors that affect the quality and reliability of the communication. Some of these factors are :

- **Encoding and decoding**: This is the process of converting the message signal into a suitable form for transmission and reception. It can involve modulation, demodulation, encryption, decryption, compression, decompression, etc. Encoding and decoding can affect the bandwidth, speed, security, and efficiency of the communication system.
- **Message**: This is the information that is communicated between the sender and the receiver. It can be a text, an image, a sound, a video, etc. The message can affect the meaning, clarity, accuracy, and completeness of the communication system.
- **Channel**: This is the medium through which the message signal travels from

Need of Modulation and Typical Applications

Modulation is the process of changing the characteristics of a carrier wave (such as amplitude, frequency or phase) according to the information contained in a message signal. Modulation is required for various reasons, such as:

- To increase the range and quality of communication: Modulation enables the transmission of low-frequency message signals over long distances by using high-frequency carrier waves, which have less attenuation and interference than low-frequency waves .
- To reduce the size and cost of the antenna: Modulation allows the use of a practical antenna, whose size is inversely proportional to the frequency

of the carrier wave. A low-frequency message signal would require a very large and expensive antenna to transmit, whereas a high-frequency carrier wave can be transmitted by a smaller and cheaper antenna .

- To enable multiplexing: Modulation allows the transmission of multiple message signals over the same channel by using different carrier waves with different frequencies, phases or amplitudes. This technique is called frequency division multiplexing (FDM), phase division multiplexing (PDM) or amplitude division multiplexing (ADM) respectively .
- To facilitate the conversion of signals: Modulation allows the inter-conversion of signals from one form to another, such as analog to digital or vice versa. Digital modulation is used to transmit digital signals over analog baseband, whereas analog modulation is used to transmit analog signals over a higher bandwidth .

Some of the typical applications of modulation are:

- Radio communication: Modulation is used to transmit radio signals, such as AM (amplitude modulation) or FM (frequency modulation) radio, over the airwaves. Radio signals are analog signals that are modulated by a carrier wave with a specific frequency range.
- Television broadcasting: Modulation is used to transmit television signals, such as analog or digital TV, over the airwaves or cable networks. Television signals are composed of video and audio signals, which are modulated by different carrier waves with different frequency bands.
- Mobile communication: Modulation is used to transmit mobile signals, such as GSM (global system for mobile communication) or CDMA (code division multiple access), over the cellular networks. Mobile signals are digital signals that are modulated by different carrier waves with different codes or frequencies.
- Internet communication: Modulation is used to transmit internet signals, such as Wi-Fi or Ethernet, over the wireless or wired networks. Internet signals are digital signals that are modulated by different carrier waves with different protocols or standards.

Fundamentals of amplitude modulation and demodulation techniques

- Amplitude modulation (AM) is a technique to transmit information via radio carrier waveform by varying the amplitude of the carrier signal in proportion to the amplitude of the modulation signal that is to be transmitted.
- Amplitude modulation can be achieved by multiplying the carrier signal with the modulation signal in the time domain or by adding the sidebands to the carrier signal in the frequency domain.
- Amplitude modulation can be classified into different types based on the

modulation index, such as double-sideband full-carrier (DSB-FC), double-sideband suppressed-carrier (DSB-SC), single-sideband (SSB), and vestigial sideband (VSB) modulation.

- Demodulation is the process of extracting the original information signal from the modulated carrier signal.
- Demodulation of AM signals can be performed by using different techniques, such as envelope detector, product detector, synchronous detector, and coherent detector .
- Envelope detector is the simplest and most common technique for demodulating AM signals. It consists of a diode, a capacitor, and a resistor. It works by rectifying the AM signal and smoothing it with a low-pass filter to recover the modulation signal.
- Product detector is a technique that multiplies the AM signal with a local oscillator signal that has the same frequency and phase as the carrier signal. It works by shifting the AM signal to the baseband and filtering out the unwanted components.
- Synchronous detector is a technique that uses a phase-locked loop (PLL) to generate a local oscillator signal that is synchronized with the carrier signal. It works by multiplying the AM signal with the PLL output and filtering out the unwanted components.
- Coherent detector is a technique that uses a reference signal that is derived from the transmitted carrier signal or a pilot signal. It works by multiplying the AM signal with the reference signal and filtering out the unwanted components.

Introduction to Wireless Communication

- Wireless communication is the transmission of voice and data without cable or wires.
- Wireless communication uses electromagnetic signals, such as radio waves, to broadcast information from sending facilities to intermediate and end-user devices .
- Wireless communication does not require a physical connection, optical fiber or other continuous guided medium for the transfer.
- Wireless communication has many advantages, such as mobility, flexibility, scalability, cost-effectiveness and reliability.
- Wireless communication has many applications, such as cellular networks, satellite systems, wireless LANs, Bluetooth, RFID, wireless sensor networks, etc.
- Wireless communication has many challenges, such as interference, noise, fading, multipath, security, power consumption, etc.
- Wireless communication has many techniques, such as modulation, coding, multiplexing, diversity, equalization, etc, to overcome the challenges and improve the performance.
- Wireless communication is a rapidly evolving field that requires a uni-

fied view of the fundamentals and the web of concepts underpinning the advances.

Overview of wireless communication

Wireless communication is a type of data communication that is performed and delivered wirelessly. This means that the devices involved in the communication do not use any physical medium, such as wires or cables, to transmit or receive information. Instead, they use electromagnetic signals, such as radio waves, microwaves, infrared, or light, to carry the information through the air or space .

Some of the advantages of wireless communication are:

- It allows mobility and flexibility for the users and devices.
- It reduces the cost and complexity of installation and maintenance of wires and cables.
- It enables the communication in remote and inaccessible areas where wired networks are not feasible or available.
- It supports a variety of applications and services, such as voice, data, video, multimedia, and Internet.

Some of the challenges of wireless communication are:

- It is susceptible to interference, noise, and fading from various sources, such as other wireless devices, weather, or obstacles.
- It has limited bandwidth and spectrum availability, which may cause congestion and performance degradation.
- It has lower security and reliability than wired networks, as the signals can be easily intercepted, jammed, or corrupted.
- It requires more power consumption and battery life for the wireless devices.

Some of the examples of wireless communication are:

- Cellular networks, such as 2G, 3G, 4G, and 5G, that provide voice and data services to mobile phones and other devices.
- Wi-Fi networks, that provide wireless Internet access to laptops, tablets, smartphones, and other devices within a local area.
- Bluetooth networks, that enable short-range wireless communication between devices, such as headphones, speakers, keyboards, mice, and printers.
- Satellite networks, that use artificial satellites orbiting the Earth to provide communication services to remote and global areas, such as television, radio, navigation, and weather.
- Radio frequency identification (RFID) networks, that use radio waves to identify and track objects, such as products, animals, or people, using RFID tags and readers.

- Wireless sensor networks, that consist of a large number of small and low-power devices, called sensors, that collect and transmit data about the physical environment, such as temperature, humidity, pressure, or motion.

Cellular Communication

Cellular communication is a type of wireless communication that uses radio waves to transmit and receive signals between mobile devices. Cellular communication is based on the concept of dividing a geographical area into small cells, each with its own base station and frequency band. Cellular communication enables multiple users to share the same frequency spectrum without interfering with each other.

Some of the topics covered in this unit are:

- **Cellular system architecture**: This topic explains the basic components and functions of a cellular system, such as mobile stations, base stations, mobile switching centers, and control channels. It also describes the different types of cells, such as macrocells, microcells, picocells, and femtocells, and their advantages and disadvantages.
- **Frequency reuse and channel assignment**: This topic explains how cellular systems use frequency reuse to increase the system capacity and reduce the interference. It also discusses the different methods of channel assignment, such as fixed, dynamic, and hybrid, and their trade-offs.
- **Interference and system capacity**: This topic analyzes the sources and effects of interference in cellular systems, such as co-channel interference, adjacent channel interference, and intermodulation interference. It also derives the expression for the system capacity and the relationship between the capacity and the cluster size.
- **Handoff and power control**: This topic explains the need and the process of handoff, which is the transfer of a mobile station from one base station to another. It also discusses the different types of handoff, such as hard, soft, and softer, and their advantages and disadvantages. It also explains the need and the methods of power control, which is the adjustment of the transmission power of the mobile stations and the base stations to maintain the signal quality and minimize the interference.
- **Cellular standards and technologies**: This topic introduces the different generations and standards of cellular systems, such as 1G, 2G, 3G, 4G, and 5G, and their features and specifications. It also describes the different technologies and protocols used in cellular systems, such as FDMA, TDMA, CDMA, GSM, GPRS, EDGE, UMTS, HSPA, LTE, and WiMAX.

Different Generations and Standards in Cellular Communication Systems

Cellular communication systems are wireless networks that use radio waves to transmit voice and data signals over a large area. Cellular communication systems have evolved over time to provide better quality, speed, capacity, and coverage for users. The different generations and standards of cellular communication systems are:

1G (First Generation Technology)

- 1G refers to the first generation of wireless cellular technology that was introduced in the late 1970s and early 1980s .
- 1G systems were voice-oriented analog cellular and cordless telephones that used frequency division multiple access (FDMA) to allocate channels.
- 1G systems had low capacity, poor voice quality, limited coverage, and no security.
- Some of the 1G standards were Advanced Mobile Phone System (AMPS), Nordic Mobile Telephone (NMT), and Total Access Communication System (TACS).

2G (Second Generation Technology)

- 2G refers to the second generation of wireless cellular technology that was commercially launched in 1991 in Finland by Radiolinja on the GSM standard.
- 2G systems were voice-oriented digital cellular and personal communication service (PCS) systems that used time division multiple access (TDMA) or code division multiple access (CDMA) to multiplex channels.
- 2G systems had higher capacity, better voice quality, wider coverage, and more security than 1G systems.
- Some of the 2G standards were Global System for Mobile Communications (GSM), Interim Standard 95 (IS-95), and Digital Enhanced Cordless Telecommunications (DECT).

3G (Third Generation Technology)

- 3G refers to the third generation of wireless cellular technology that was first launched in 2001 in Japan by NTT DoCoMo on the W-CDMA standard.
- 3G systems were data-oriented cellular and wireless wide area network (WAN) systems that used wideband CDMA (W-CDMA) or CDMA2000 to increase bandwidth and data rates.
- 3G systems had higher speed, capacity, and quality than 2G systems, and supported multimedia services, such as video, audio, and internet access.

- Some of the 3G standards were Universal Mobile Telecommunications System (UMTS), CDMA2000, and High-Speed Packet Access (HSPA).

4G (Fourth Generation Technology)

- 4G refers to the fourth generation of wireless cellular technology that was first launched in 2009 in Norway and Sweden by TeliaSonera on the LTE standard.
- 4G systems were data-oriented cellular and wireless LAN (WLAN) systems that used orthogonal frequency division multiple access (OFDMA) or single carrier FDMA (SC-FDMA) to achieve higher data rates and lower latency.
- 4G systems had higher speed, capacity, and quality than 3G systems, and supported broadband services, such as video streaming, online gaming, and cloud computing.
- Some of the 4G standards were Long Term Evolution (LTE), WiMAX, and LTE-Advanced.

5G (Fifth Generation Technology)

- 5G refers to the fifth generation of wireless cellular technology that was first launched in 2018 in the USA by Verizon and AT&T on the mmWave standard .
- 5G systems are data-oriented cellular and wireless LAN (WLAN) systems that use massive multiple-input multiple-output (MIMO), beamforming, and millimeter wave (mmWave) technologies to achieve higher data rates, lower latency, and more reliability.
- 5G systems have higher speed, capacity, and quality than 4G systems, and support ultra-reliable low-latency communication (URLLC), massive machine-type communication (mMTC), and enhanced mobile broadband (eMBB) services.
- Some of the 5G standards are New Radio (NR), 5G NR, and 5G NR-U.

Fundamentals of Satellite & Radar Communication

Satellite Communication

- Satellite communication is the transmission of signals between a satellite and the earth stations or other satellites.
- Satellites are relay stations in space that receive, amplify, and retransmit signals to different locations on the earth or to other satellites.
- Satellites can provide global coverage, high bandwidth, and reliability for voice, video, and data communications.

- Satellites operate in different orbits, such as geostationary orbit (GEO), medium earth orbit (MEO), low earth orbit (LEO), and highly elliptical orbit (HEO).
- Satellites use different frequency bands, such as C-band, Ku-band, Ka-band, and L-band, for uplink and downlink transmissions.
- Satellites have different components, such as antennas, transponders, solar panels, batteries, and attitude control systems, that enable them to function in space.
- Satellites have different applications, such as direct broadcast, high throughput, VSAT, mobile, navigation, remote sensing, and scientific exploration.

Radar Communication

- Radar communication is the detection and ranging of objects using radio waves.
- Radar systems consist of a transmitter, a receiver, an antenna, and a display unit.
- Radar systems emit electromagnetic pulses that reflect off the target objects and return to the receiver.
- Radar systems measure the time delay, frequency shift, and angle of arrival of the reflected pulses to determine the distance, speed, and direction of the target objects.
- Radar systems use different types of modulation, such as pulse modulation, frequency modulation, and phase modulation, to encode information on the transmitted pulses.
- Radar systems use different types of antennas, such as parabolic, horn, phased array, and slot, to radiate and receive the pulses in different directions and patterns.
- Radar systems have different applications, such as air traffic control, navigation, surveillance, weather forecasting, and military defense.